

WaterQuest

SOLUTIONS

Version 1.0 · October 2026

Valves

The valve arrangement defines the machine behaviour.

Choose the hydraulic function before choosing a size: direction, flow regulation, pressure control, holding, isolation or protection. The collection includes twenty valve/accessory reference documents, with source specifications, curves, ordering information and available drawings.

Valve / accessory reference	Length	Page
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Design & environmental advantages

Water-specific materials and sealing arrangements support low-viscosity service. Matching function, actuation and interface makes a coherent circuit while avoiding hydraulic oil in the water circuit.

Efficiency and selection

Valve losses are part of the system energy balance. At a given flow, avoid unnecessary differential pressure. Pressure regulation and throttling reject heat; unloading or control changes may reduce idle losses.

Define the quotation and technical submittal

Provide the circuit, duty flow, supply/return/pilot pressure, normal and failure states, leakage requirement, electrical supply, fluid, seals, port standard and mounting. Ask for selected code, curve at duty, wiring and current approval evidence where applicable.

Source checks that affect selection

Relief DN12 capacity is 0-120 L/min in the manufacturer table. Its chart has unequal numeric flow steps at equal spacing: read labelled points, not horizontal slope. Component and historical certificate references do not certify the assembled machine.

Each following model section is also supplied as its own PDF. Model identity, source conditions and technical references travel with the standalone sheet. Send the operating duty to bruce@waterquestsolutions.com for configuration selection.

Technical source editions and unresolved discrepancies are listed in the included verification record. Local brochure edition: 17 Oct 2026.

2-2 Directional Valve

Selection data & performance | Manufacturer technical reference.

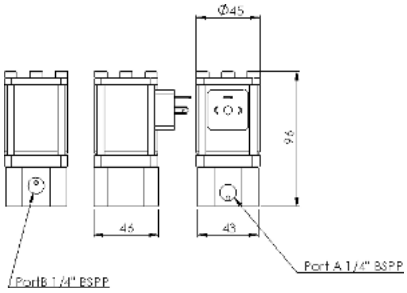
Design

Available in both normally open and normally closed configuration, the 2/2 valve is supplied as a cartridge design and therefore can be arranged to offer multiple functions from a single block; reducing cost on expensive fittings and pipe work. Please contact TWHC for manifold options or cavity details.

The wet pin design solenoid actuator has no dynamic sealing and therefore produces adequate force to directly actuate the valve with an orifice size of 2.5mm up to a maximum working pressure of 160 bar. For lower actuation pressures, the diameter of the flow gallery can be increased to deliver a much greater cross section of flow path for higher flow conditions.
For higher flow capacity, please see data sheet for the 2/2 pilot valves.

SPECIFICATION	
Max Inlet Pressure	200 Bar
Operating Pressure Range	8 to 160 Bar
Flow Range	0-10 L/min
Flow Characteristics	Cv Rate 0.15 Kv Rate 0.13
Feed Gallery Diameter	2.5mm
Porting	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer
Solenoid Operation	12 & 24V DC
Media	Water Filtered to 25µm
Temperature Range	1 to 50°C





Note: Porting orientation changes with spring condition

ORDERING CODES	SPRING POSITIONS	12V DC	24V DC
 B In	N/C	226AASW	226AOSW
 A In	N/O	221AASW	221AOSW

Specify the required normal state, actuation supply and flow direction. Verify the electrical load, mounting pattern and pressure differential at switching. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Directional Valve V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

2-2 Pilot Valves

Selection data & performance | Manufacturer technical reference.

Design

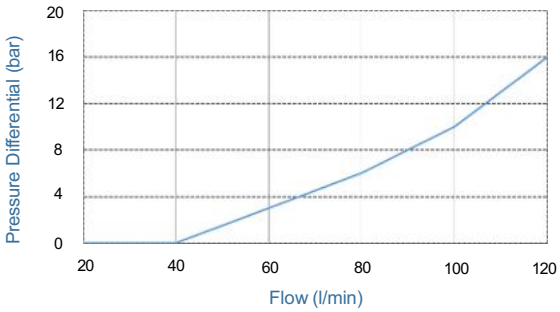
Available in both normally open and normally closed configuration, the 2/2 valve is supplied as a manifold design and therefore can be arranged to offer multiple functions from a single block; reducing cost on expensive fittings and pipe work. Please contact TWHC for manifold options or cavity details.

The pilot section wet pin design solenoid actuator has no dynamic sealing and therefore produces adequate force to directly actuate the valve with an orifice size of 2.5mm up to a maximum working pressure of 160 bar. For lower flow capacity, please see data sheet for the 2/2 directional flow control.

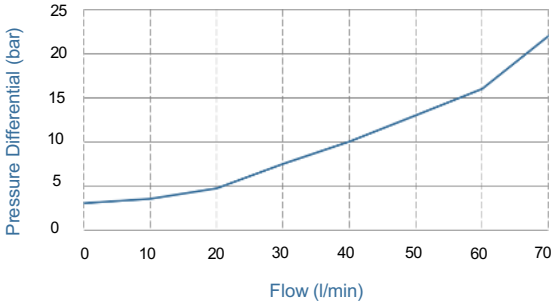
SPECIFICATION		
	DN9	DN25
Max Inlet Pressure	200 Bar	200 Bar
Operating Pressure Range	8 to 160 Bar	8 to 160 Bar
Flow Range	0-60 L/min	0-120L/min
Flow Characteristics	Cv Rate 0.15 Kv Rate 0.17	Cv Rate 2.5 Kv Rate 2.87
Feed Gallery Diameter	9mm	20mm
Porting	3/8" BSPP	1" BSPP
Construction Materials	A.I.S.I 316/Polymer	A.I.S.I 316/Polymer
Solenoid Operation	12 & 24V DC	12 & 24V DC
Media	Water Filtered to 25µm	Water Filtered to 25µm
Temperature Range	1 to 50°C	1 to 50°C

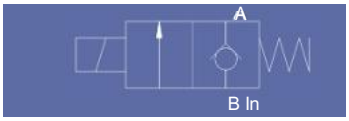
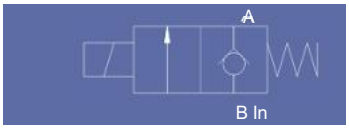
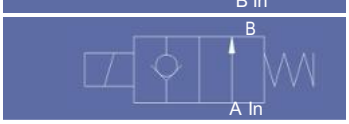


DN25 Fully opened pressure drop against flow rates



DN9 Fully opened pressure drop against flow rates



ORDERING CODES	SPRING POSITIONS	PORT ORIFICE	12V DC STAND ALONE	12V DC MANIFOLD MOUNT	24V DC STAND ALONE	24V DC MANIFOLD MOUNT
	N/C	DN9	226CASW	2M26CASW	226COSW	2M26COSW
	N/O	DN9	221CASW	2M21CASW	221COSW	2M21COSW
	N/C	DN25	226FASW	2M26FASW	226FOSW	2M26FOSW
	N/O	DN25	221FASW	2M21FASW	221FOSW	2M21FOSW

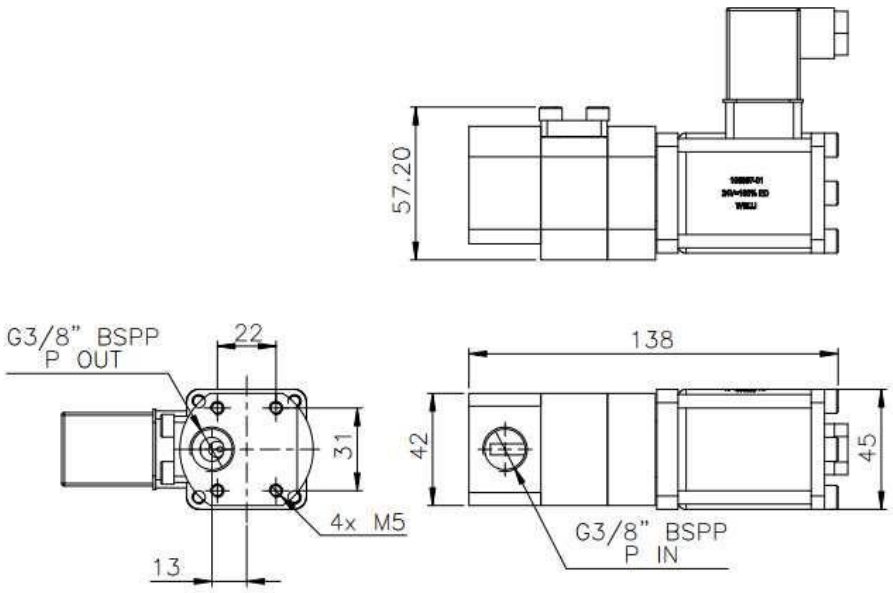
Select the required pilot pressure, fail position, flow direction and DN size. Verify pressure drop at full flow and the pilot arrangement before ordering. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Pilot Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

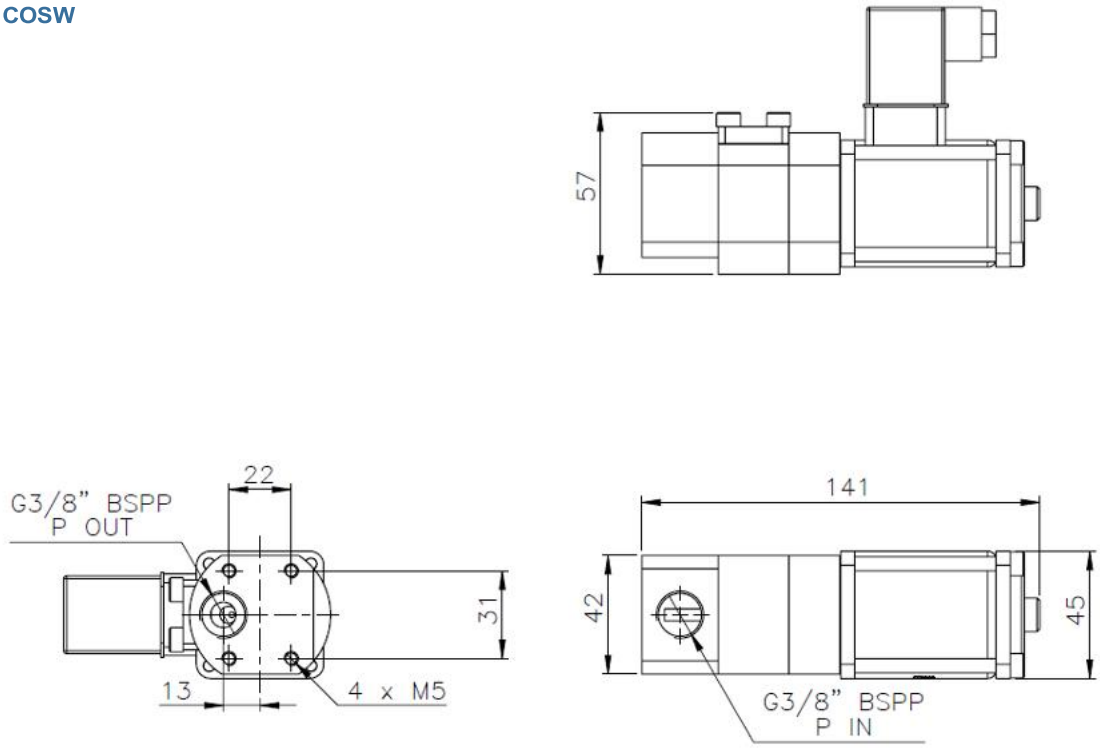
2-2 Pilot Valves

Dimensions, connections & configuration | Manufacturer technical reference.

226COSW



221COSW



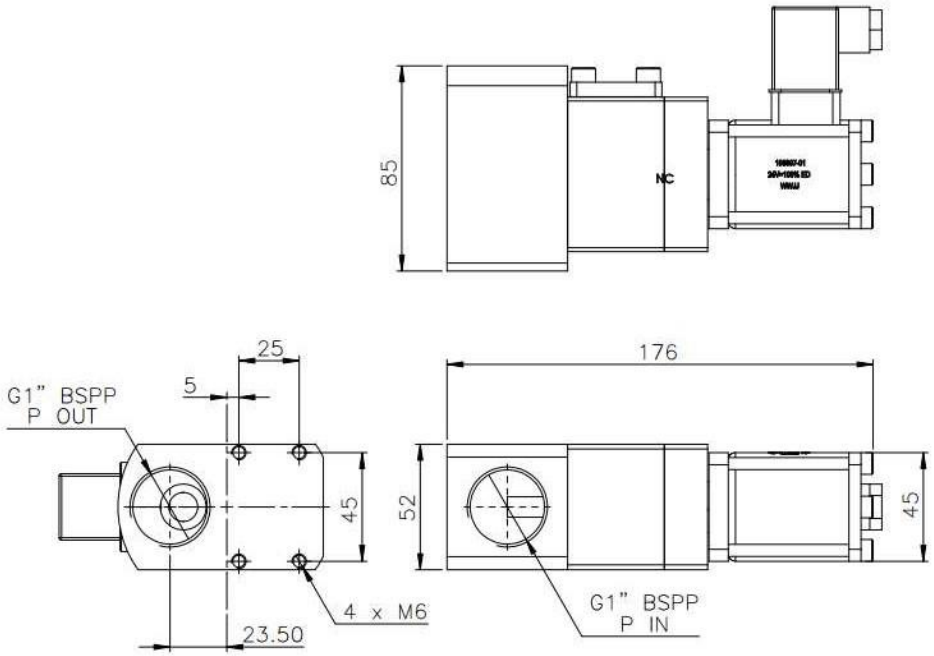
Select the required pilot pressure, fail position, flow direction and DN size. Verify pressure drop at full flow and the pilot arrangement before ordering. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Pilot Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

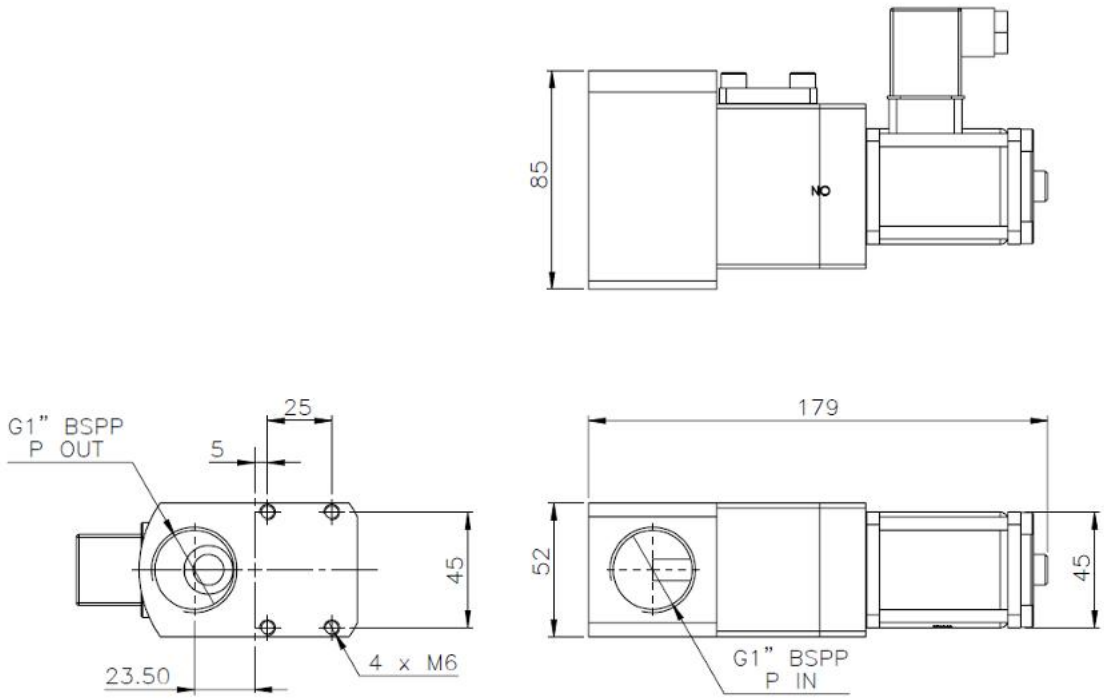
2-2 Pilot Valves

Dimensions, connections & configuration | Manufacturer technical reference.

226FOSW



221FOSW



Select the required pilot pressure, fail position, flow direction and DN size. Verify pressure drop at full flow and the pilot arrangement before ordering. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Pilot Valves V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

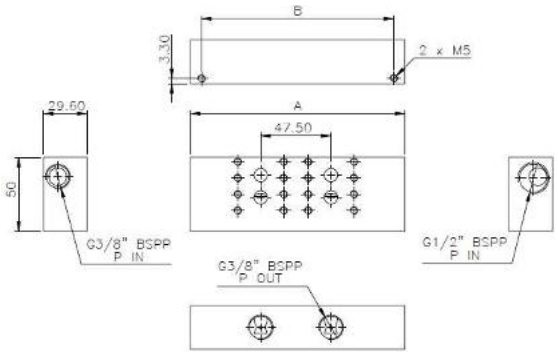
2-2 Pilot Valves

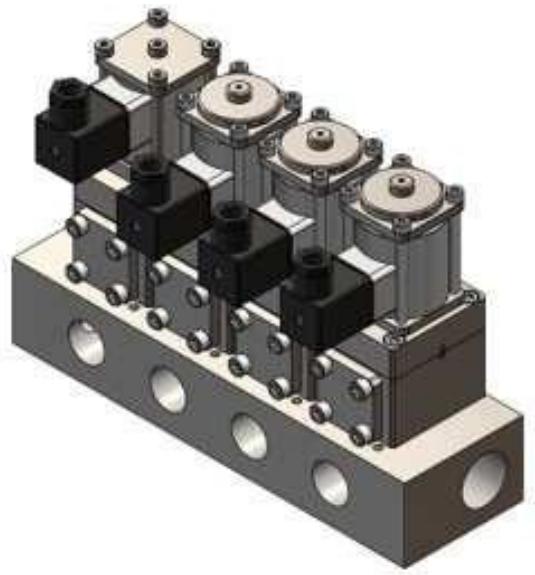
Dimensions, connections & configuration | Manufacturer technical reference.

Available in 1 to 5 stations and produced in 316 stainless steel. The manifolds will accept the 2/2 normally open and normally closed valve configurations in all actuation formations such as solenoid or pilot

DN9

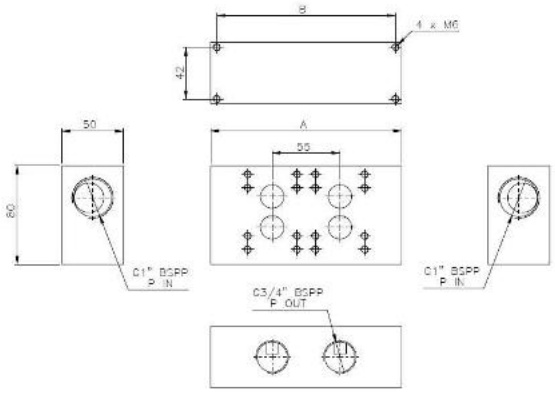
NO. OF STATIONS	1	2	3	4	5
Dimension A	95	145	195	245	295
Dimension B	80	130	180	230	280





DN25

NO. OF STATIONS	1	2	3	4	5
Dimension A	100	155	210	265	320
Dimension B	90	145	200	255	310



Assembly Layout Example

Consisting of:

- 500-026: 4 Station Manifold
- 3 x 2M26FOSW: DN25 Manifold Mounted 2/2 Normally Closed Valve
- 2M21FOSW: DN25 Manifold Mounted 2/2 Normally Open Valve

ORDERING CODE	1 STATION	2 STATION	3 STATION	4 STATION	5 STATION
DN9	500-009	500-010	500-011	500-012	500-005
DN25	500-023	500-024	500-025	500-026	500-027

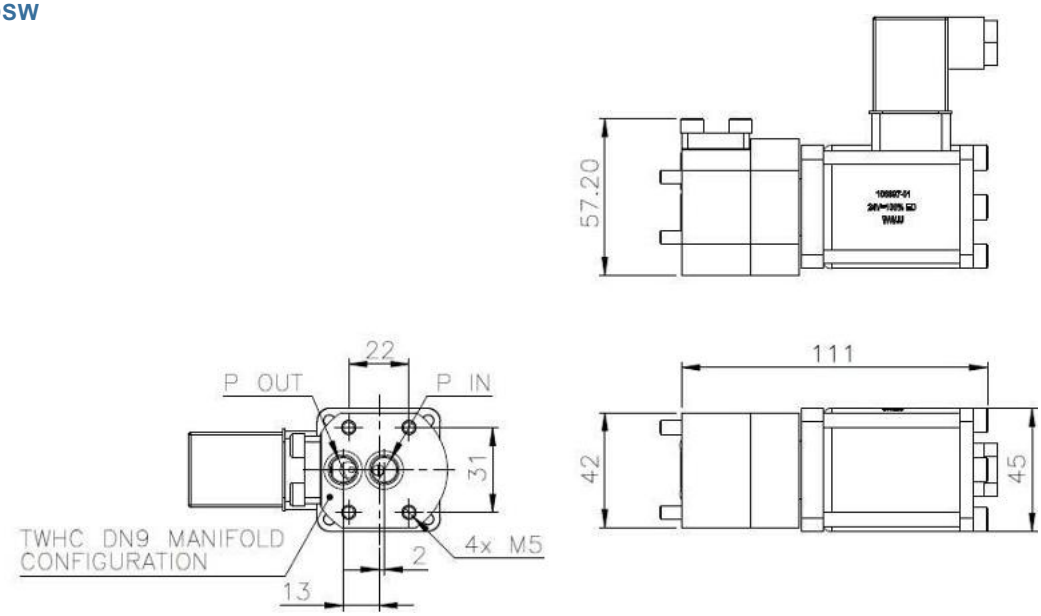
Select the required pilot pressure, fail position, flow direction and DN size. Verify pressure drop at full flow and the pilot arrangement before ordering. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Pilot Valves V.01.25.pdf, p. 4. Manufacturer labels and dimensions retained.

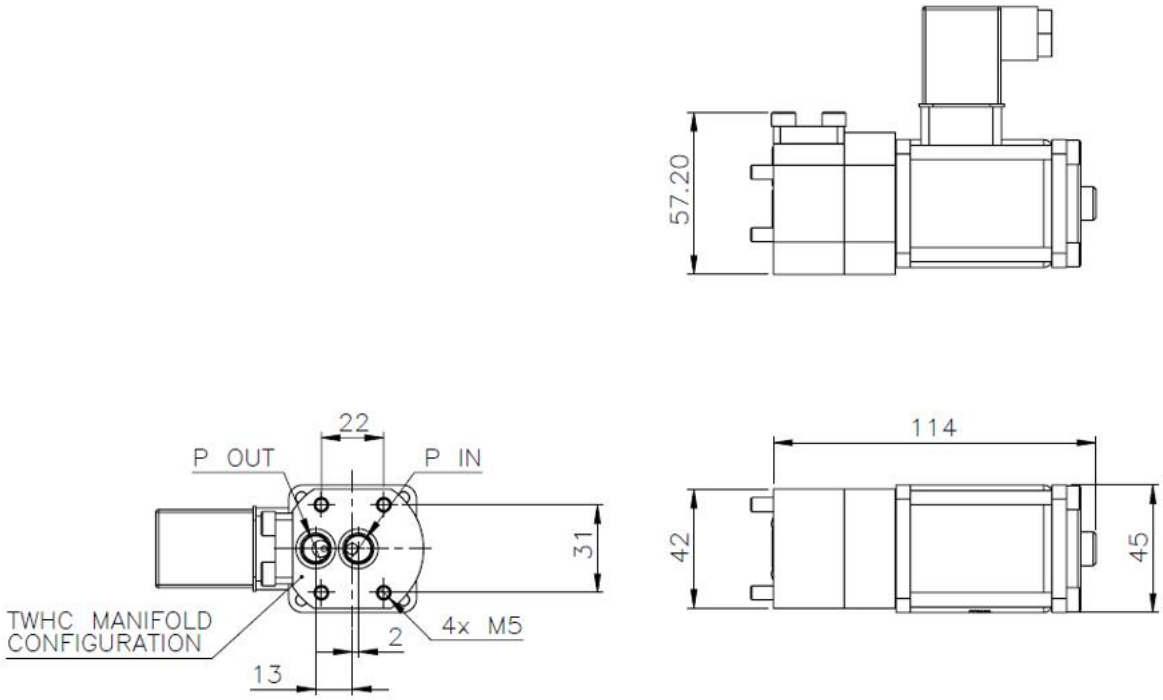
2-2 Pilot Valves

Dimensions, connections & configuration | Manufacturer technical reference.

2M26COSW



2M21COSW



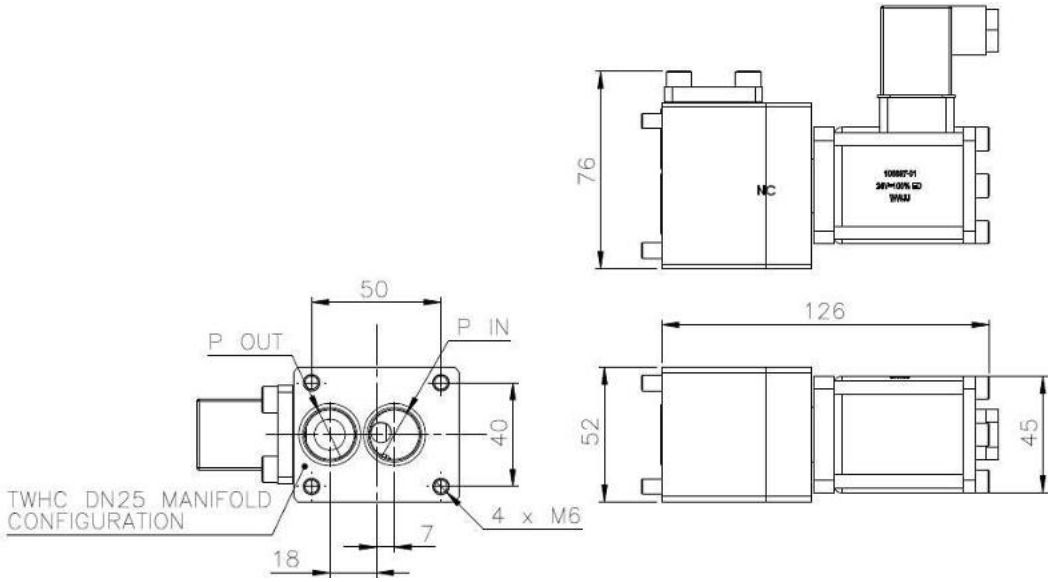
Select the required pilot pressure, fail position, flow direction and DN size. Verify pressure drop at full flow and the pilot arrangement before ordering. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Pilot Valves V.01.25.pdf, p. 5. Manufacturer labels and dimensions retained.

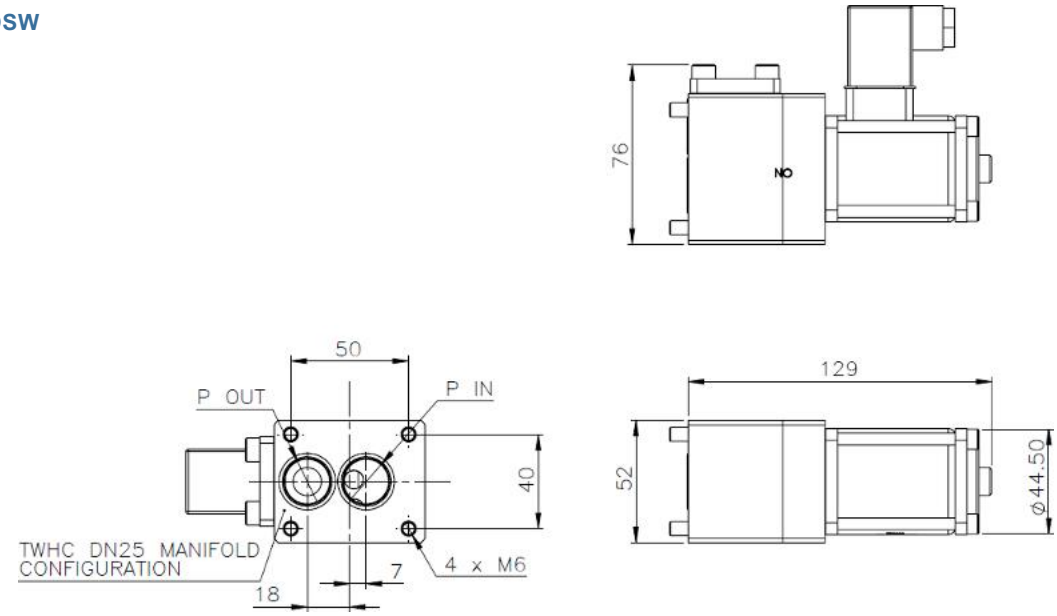
2-2 Pilot Valves

Dimensions, connections & configuration | Manufacturer technical reference.

2M26FOSW



2M21FOSW



Select the required pilot pressure, fail position, flow direction and DN size. Verify pressure drop at full flow and the pilot arrangement before ordering. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

2-2 Pilot Valves V.01.25.pdf, p. 6. Manufacturer labels and dimensions retained.

4-2 & 3-2 Directional Control Valves

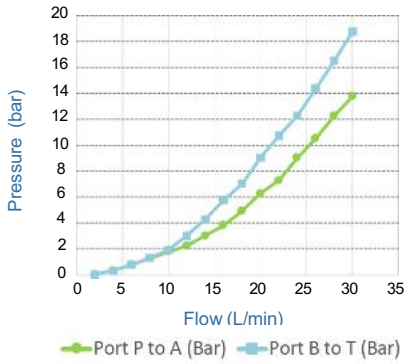
Selection data & performance | Manufacturer technical reference.

A patented range of valves which control the directional flow of corrosive fluids in hydraulic systems. Designed specifically for fluids with low viscosity, the valves offer high efficiency with minimal energy input for actuation.
Available in 4/2 & 3/2 control functions, the rest position in both normally open or normally closed are also offered. Various porting configurations including stand alone (BSPP), TWHC manifold or Cetop 3 are available. Please see manifolds data sheet for options and cavity details.

SPECIFICATION	
Max Inlet Pressure	160 Bar
Operating Pressure Range	0 to 160 Bar
Flow Range	0-30 L/min
Flow Characteristics	Cv Rate 0.62 Kv Rate 0.75
Feed Gallery Diameter	6mm
Porting	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer
Solenoid Operation	24V DC
Media	Water
Electrical Connections	Spade or Hirschmann



Flow Characteristics



PWM Circuit

For direct acting solenoid actuation above 20 bar, actuation card (part number: 004-010) must be used. This minimises the electrical power consumption whilst at the same time maximising the actuation force of the solenoid. For power requirement and wiring instructions please see actuation card data sheet.

Actuation

Solenoid actuation has a manual over ride as standard. Actuation options available are hand, detent, air pilot, water pilot and oil pilot. We also offer a combination of actuation and return; please contact TWHC for further information.

ORDERING CODES	STANDARD BUILD	MANIFOLD MOUNT	CETOP MOUNT
 4/2 Valve	201BBSW	2M01BBSW	2C01BBSW
 3/2 Valve Spring to Open P	208BBSW	2M08BBSW	2C08BBSW
 3/2 Valve Spring to Closed P	203BBSW	2M03BBSW	2C03BBSW

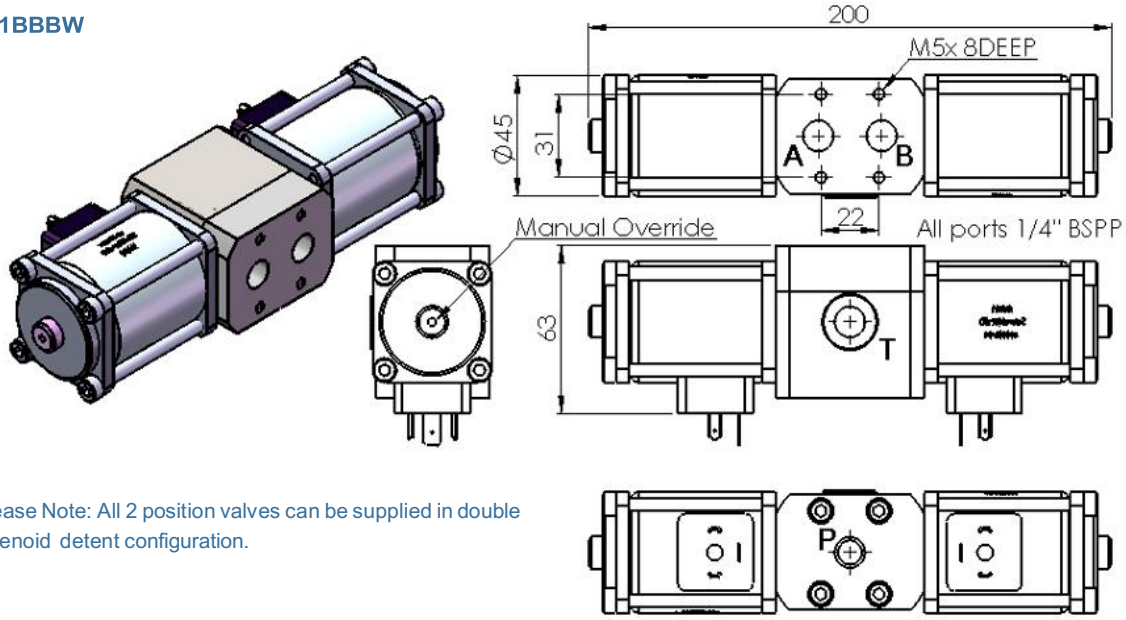
Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

4-2 & 3-2 Directional Control Valves

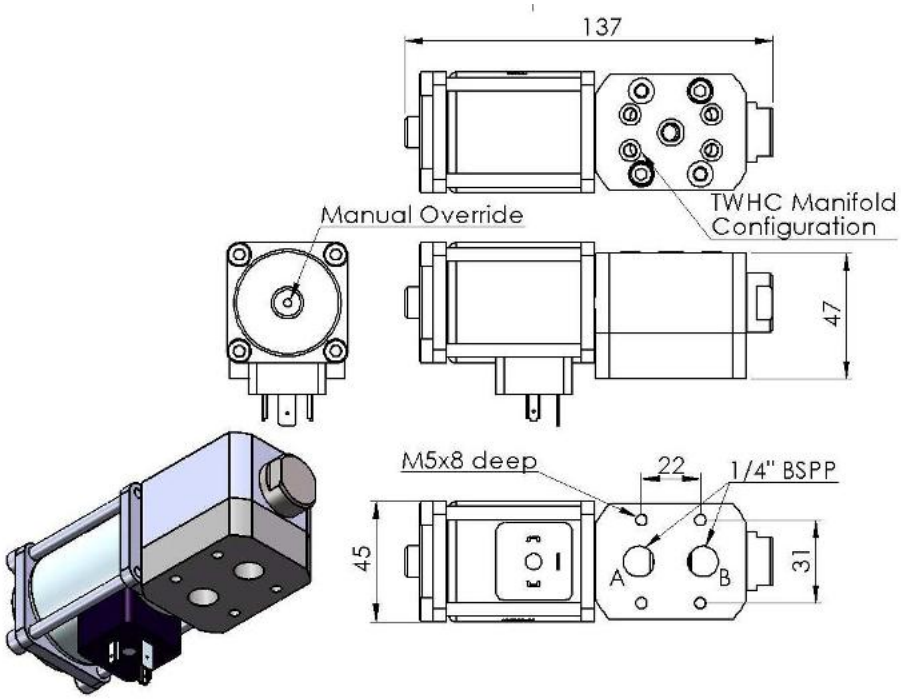
Dimensions, connections & configuration | Manufacturer technical reference.

201BBBW



Please Note: All 2 position valves can be supplied in double solenoid detent configuration.

2M01BBSW



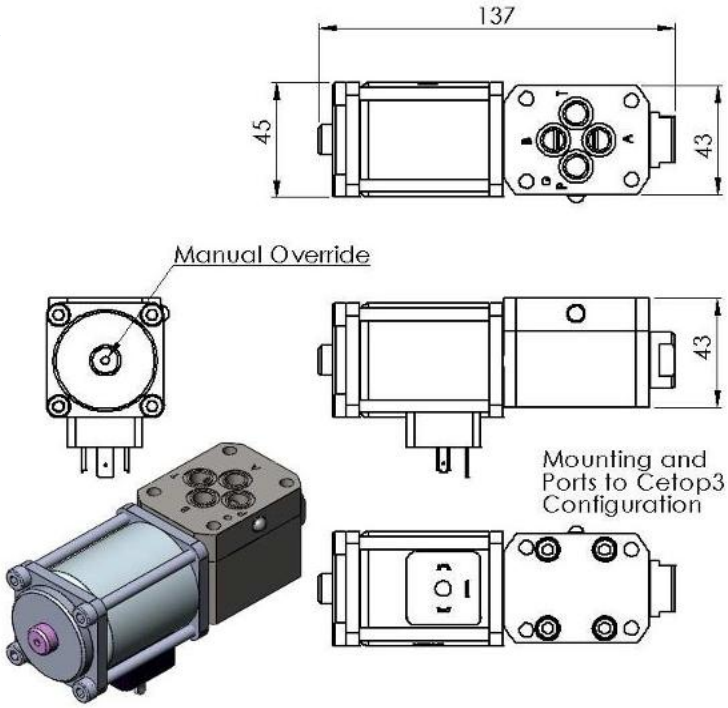
Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

4-2 & 3-2 Directional Control Valves

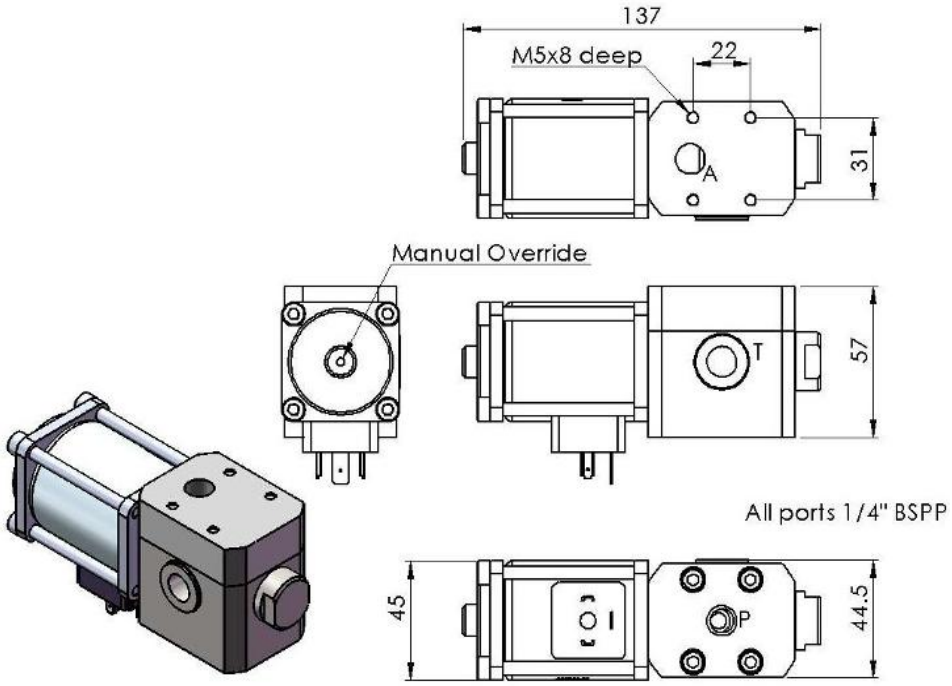
Dimensions, connections & configuration | Manufacturer technical reference.

2C01BBSW, 2C03BBSW & 2C08BBSW



Please Note: The sprung position is port P joined to port B, port A joined to port T.

208BBSW



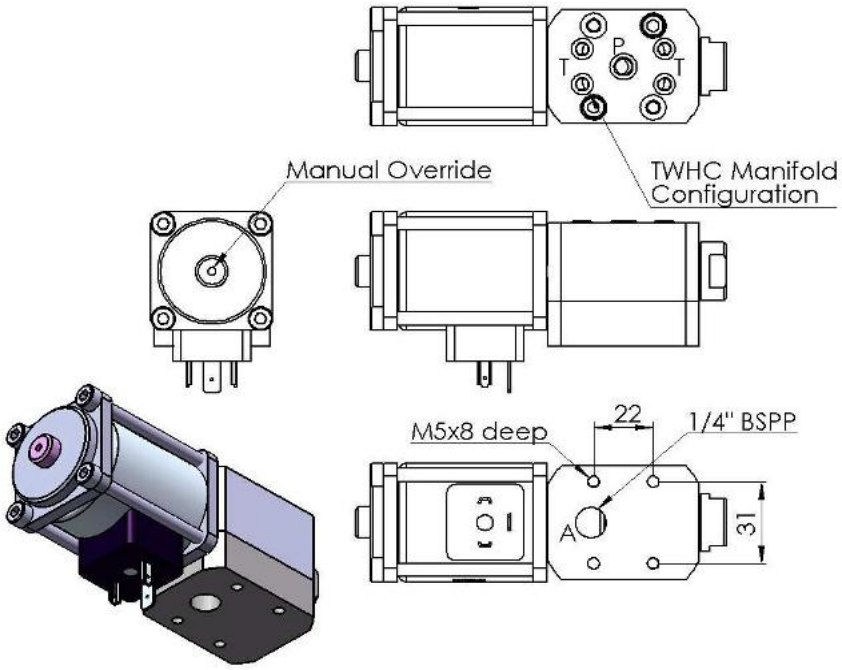
Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

4-2 & 3-2 Directional Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

2M08BBSW



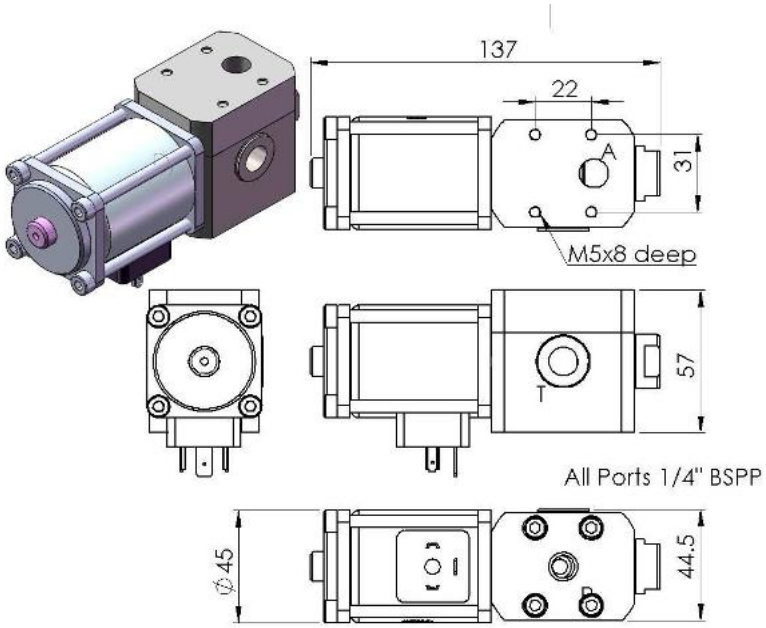
Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 4. Manufacturer labels and dimensions retained.

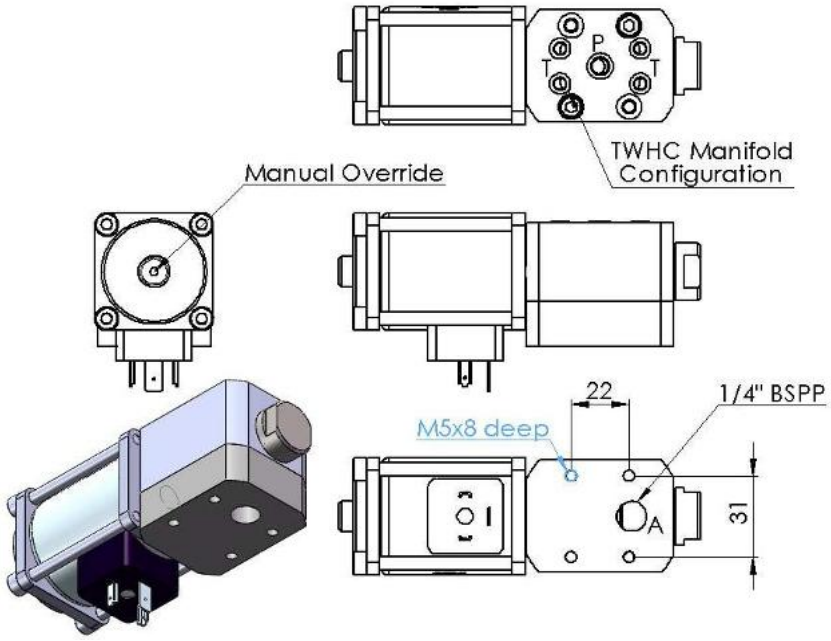
4-2 & 3-2 Directional Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

203BBSW



2M03BBSW



Please Note: All system components must be manufactured in non-corrosive materials; i.e. Stainless Steel or plastic where suitable. Ferrous particles & oxides are very aggressive and will seriously damage the health of your Water Hydraulics product. High wear; erratic performance and premature failure will result if this is not adhered to.

Match the spool function, actuator flow and switching sequence. Include return pressure and define the state required after loss of power. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-2 & 3-2 Directional Control Valves V.01.25.pdf, p. 5. Manufacturer labels and dimensions retained.

4-3 Directional Control Valves

Selection data & performance | Manufacturer technical reference.

A patented range of valves which control the directional flow of corrosive fluids in hydraulic systems. Designed For operation on tap water to EU directive 98/83/EC. The valves offer high efficiency with minimal energy input for actuation. Also available in 4/2 & 3/2 control functions; please see data sheet for further information. Various porting configurations including stand alone (BSPP), TWHC manifold or CETOP 3 are available. Please see manifolds data sheet for options and cavity details.

SPECIFICATION	
Max Inlet Pressure	200 Bar
Operating Pressure Range	0 to 160 Bar
Flow Range	0-30 L/min
Flow Characteristics	Cv Rate 0.62 Kv Rate 0.75
Feed Gallery Diameter	6mm
Porting	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer
Solenoid Operation	24V DC
Media	Water
Electrical Connections	Spade or Hirschmann



PWM Circuit

For solenoid actuation above 20 bar, actuation card (part number: 004-010) must be used. This minimises the electrical power consumption whilst at the same time maximising the actuation force of the solenoid. For power requirement and wiring instructions please see actuation card data sheet.

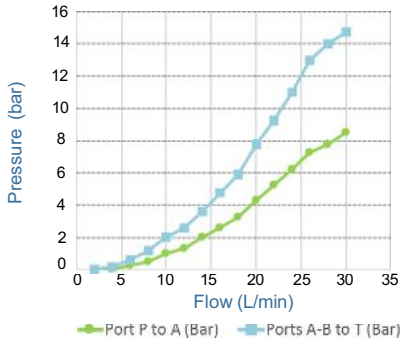
Actuation

Solenoid actuation has a manual over ride as standard. Actuation options available are hand, detent, air pilot, water pilot and oil pilot. We also offer a combination of actuation and return; please contact TWHC for further information.

Centre Position

A full all ports blocked at the centre position is not possible with this size of valve. The valves are not offered for 100% seal performance. If load holding or leak free performance is required then the use of a PO Check valve to mount directly onto the A & B port face is advised.

Flow Characteristics



ORDERING CODES	STANDARD BUILD	MANIFOLD MOUNT	CETOP MOUNT
 Centre Position Closed P AB&T connected	206BBSW	2M06BBSW	2C06BBSW
 Centre Position All Ports connected	212BBSW	2M12BBSW	2C12BBSW

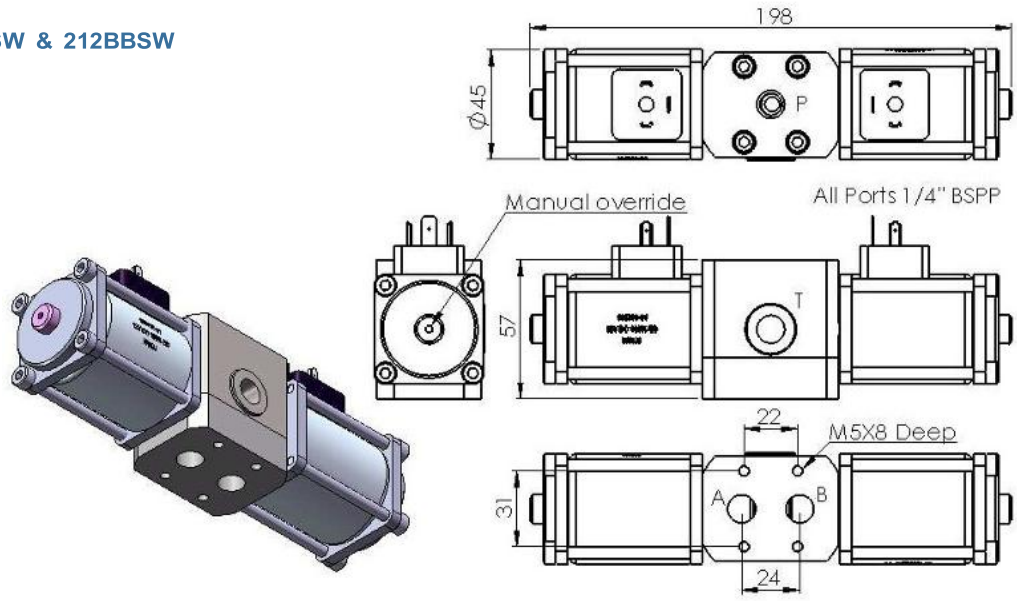
Specify the centre condition and the load response in neutral. Check pressure drop, coil supply, mounting and external load-holding needs. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-3 Directional Control Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

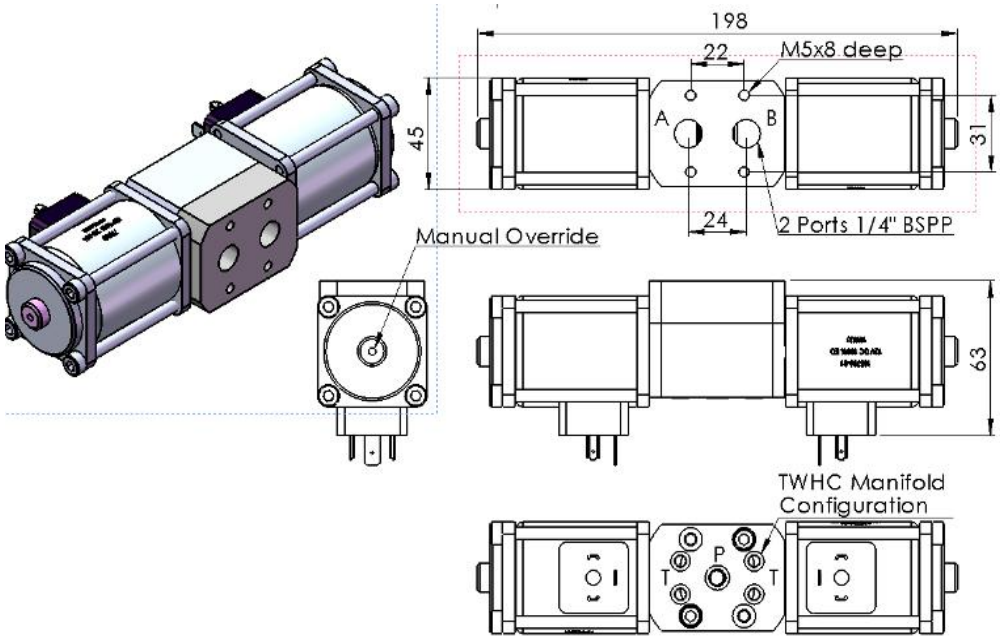
4-3 Directional Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

206BBSW & 212BBSW



2M06BBSW & 2M12BBSW



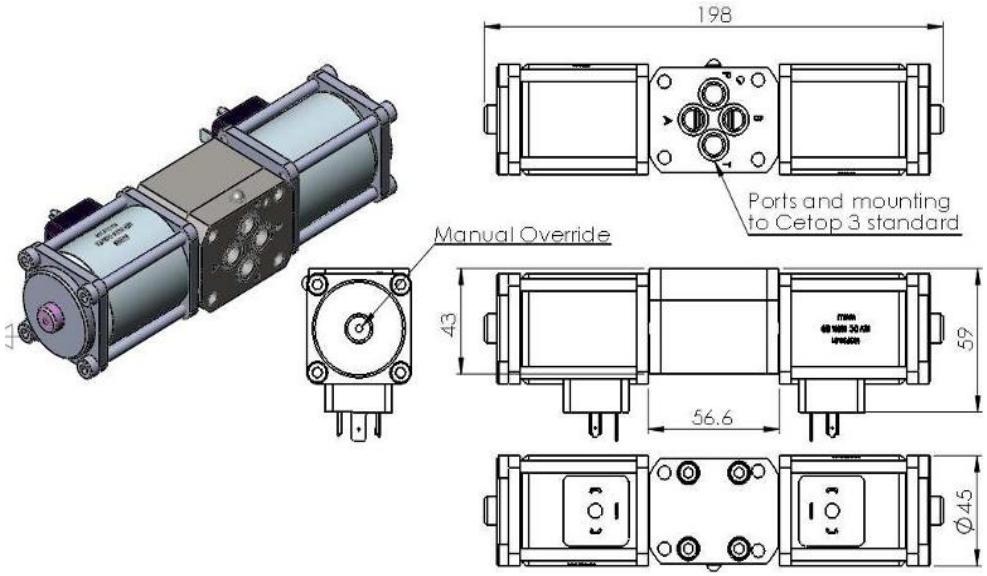
Specify the centre condition and the load response in neutral. Check pressure drop, coil supply, mounting and external load-holding needs. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-3 Directional Control Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

4-3 Directional Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

2C06BBSW & 2C12BBSW



Specify the centre condition and the load response in neutral. Check pressure drop, coil supply, mounting and external load-holding needs. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

4-3 Directional Control Valves V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

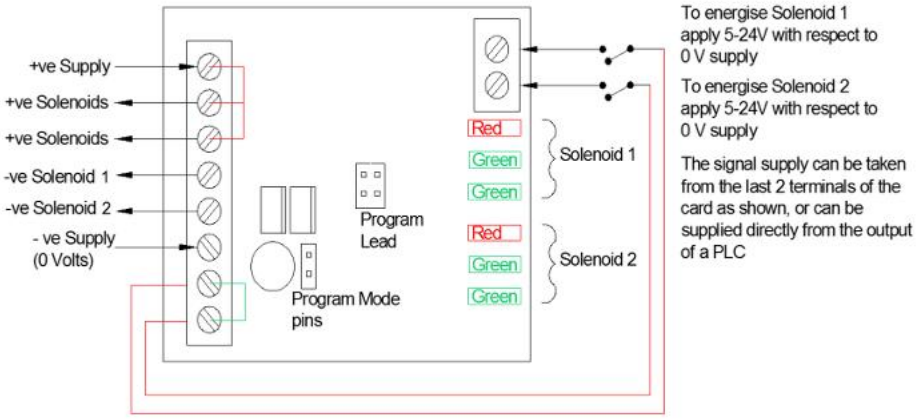
Actuation Cards

Selection data & performance | Manufacturer technical reference.

The directional control valves are supplied with a 12V DC coil*. The PCB will control the duration of supply to ensure the maximum temperature of the coil is not exceeded. The solenoid should not be switched on/off continuously with a duty cycle shorter than 1 second.

Power supply

A power supply capable of supplying a regulated 24V DC (supply) at 5 amps is required to the actuation card at all times.
The schematic below shows the signal supply can be derived from the last 2 terminals of the board; alternatively the signal can be supplied directly from a PLC (6mA if the output from the PLC is 24V or 1mA if the output is 5V).



With the connections made as shown, by supplying a 5-24V signal to the input a sequence will commence. The solenoid will be supplied with an initial 24V supply for 100ms. The average voltage the solenoid will then see is 15V for a further 2 seconds. A further reduction to 10V is then made to reduce the temperature generated in the solenoid under hold conditions; this will also reduce the power consumption of the system.
The sequence is illustrated by the LED's; the red is the holding condition.

Please note: once energised, the card will remain in the holding condition with a signal voltage of only 1.3V, therefore care should be taken when routing the signal cables to avoid interference which may induce voltages high enough to trigger the signal.

Temperature

The rated operating range for this module is –40°C to 50°C ambient.

Mounting

The board can be supplied with optional DIN rail mounting as shown below.

Din Rail Mounting





Match card type, control signal and electrical supply to the selected valve actuator. Specify enclosure, wiring, interlocks and commissioning requirements. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Actuation Cards V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

Actuation Cards

ORDERING CODES	24V	48V**	DIMENSIONS
Actuation Card Only	004-010	004-013	H 38mm L 78mm W 42mm
1 Card including Din Rail Mounting	004-104	004-114	H50mm L 80mm W 45mm
2 Cards including Din Rail Mounting	004-105	004-115	H 50mm L 80mm W 90mm
3 Cards including Din Rail Mounting	004-106	004-116	H 50mm L 80mm W 136mm
Mounting Only (1 Card)	004-101		H 50mm L 80mm W 45mm
Mounting Only (2 Cards)	004-102		H 50mm L 80mm W 90mm
Mounting Only (3 Card)	004-103		H 50mm L 80mm W 136mm

*Specially programmed versions available for 28v automotive applications
**48v coil version requires 24v and 48v 2.5A PSU

Match card type, control signal and electrical supply to the selected valve actuator. Specify enclosure, wiring, interlocks and commissioning requirements. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Actuation Cards V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

Check Valves

Selection data & performance | Manufacturer technical reference.

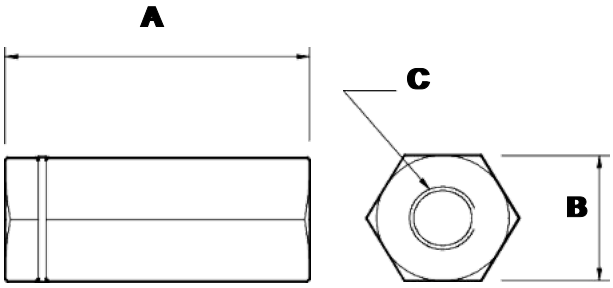
The check valves are a unique range of valves which are supplied in 316 stainless steel and high grade engineering polymer as standard, for operation on tap water to EU directive 98/83/EC. Available in 3 sizes, the range offers consistent performance even after lengthy periods of inactivity.

SPECIFICATION					
	DN6	DN9	DN12	DN16	DN25
Pressure Range (bar)	0.1 - 200	0.1 - 200	0.1 - 200	0.1 - 200	0.1-200
Feed Gallery Diameter	6	9.5	12.7	16	25
Cv	0.65	0.65	1.92	4.5	6.6
Max Flow rate L/min	30	50	60	100	200
"A" Length	56.5	89	70	147.3	147.3
"B" A/F	25.6	22.2	29	42.4	42.4
"C" Thread	1/4" BSPP	3/8" BSPP	1/2" BSPP	3/4" BSPP	1" BSPP

The sealing and control of flow occurs at 2 separate positions within the valve. This ensures the seals are not subjected to high fluid velocities minimising the potential of seal flow erosion.

Cracking Pressure

Standard valves are offered with 1 bar cracking pressure. Cracking pressures from 0.1 bar to 6 bar are available. If specific pressures are required then please consult TWHC.



Dimensions in mm (see table)



ORDERING CODES	6	DN9	DN12	DN16	DN25
Check	SW	210CXSW	210DXSW	210EXSW	210FXSW

Specify cracking pressure, flow direction, pressure drop and allowable reverse leakage. Check the installation orientation and port/interface details. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Check Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

DC Microstep Drive with Q Programming (ST5-Q-NN)

Selection data & performance | Manufacturer technical reference.

Product Features

FEATURES

- Programmable microstepping drive with advanced current control
- Q Programmer™ for robust motion control programming
- 744 lines of stored program capability
- Math calculations using analog and digital parameters
- Supports all "S" drive control modes as well
- Modbus/RTU for easy PLC and HMI communication
- Compatible with many 3rd party HMIs
- Wide current range 0.1 to 5.0 A/phase (peak of sine) with idle current reduction
- Advanced anti-resonance algorithm
- Torque ripple smoothing
- 8 digital inputs, 4 digital outputs, optically isolated
- 2 analog inputs, +/-10 volt range
- RS-232 cable and mating connectors included



DESCRIPTION

The ST5-Q-NN stepper drive is a DC-powered microstepping drive for controlling two-phase, bipolar step motors. It offers advanced current control and a sophisticated 3rd generation anti-resonance algorithm that electronically dampens motor and system resonances to improve motor smoothness and usable torque over a wide speed range. The drive also employs electronic torque ripple smoothing and microstep emulation to greatly reduce motor noise and vibration. The drive must be powered from 24-48 VDC and can output up to 5.0 A/phase (peak-of-sine) to the step motor. Over-voltage, over-temperature and over-current protection features prevent damage while running in adverse conditions. The drive is complemented by a specifically matched set of NEMA 11 through NEMA 23 frame stepper motors (see Related and Recommended products below).

The ST5-Q-NN can operate in all of the same control modes as the S model (pulse & direction, velocity, streaming commands, SiNet Hub operation), plus it has the ability to run stand-alone Q programs stored in non-volatile memory. Q programs are created using the Q Programmer™ software, which provides multi-tasking, math functions, conditional processing, data register manipulation, and more features in a robust yet simple text-based programming language. The drive is setup and configured using Applied Motion's ST Configurator™ software. Preconfigured motor setup files included with ST Configurator™ make it easy to set up the drive for optimum results.

For connecting to external devices such as control signals, incremental encoders, limit switches, proximity or photoelectric sensors, PLC I/O, lamps, and other devices, the drive comes with 8 digital inputs, 4 digital outputs, and 2 single-ended analog inputs (analog inputs can be wired together as 1 differential analog input). Adjustable digital filters are present on the digital inputs for enhanced reliability in noisy environments.

The drive comes with an RS-232 port for configuration, serial communications and programming.

All ST drives are CE approved and RoHS compliant.

Match motor current, voltage, command interface and operating profile. Specify wiring, enclosure and interlocks; this is an electrical accessory, not a fluid valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

DC Microstep Drive with Q Programming (ST5-Q-NN)

Dimensions, connections & configuration | Manufacturer technical reference.

SPECIFICATIONS	
Model Number	ST5-Q-NN
Part Number	5000-128
Supply Voltage	24-48 VDC
Supply Voltage Type	DC
Control Modes	· Step & Direction · Velocity (Oscillator) · Streaming Commands · Analog Positioning · Encoder Following · Q Programming · SiNet Hub Compatible · Modbus
Output Current	0.1-5.0 A/Phase
Communication Ports	RS-232
Encoder Feedback	No
Step Resolution	· Full · Half · Microstepping · Microstep Emulation
Idle Current Reduction	0-90%
Setup Method	Software setup
Digital Inputs	8
Digital Outputs	4
Analog Inputs	1 differential or 2 single-ended
Dimensions	5.0 x 3.0 x 1.75 inches
Weight	10.4 oz
Operating Temperature Range	0-70 °C
Ambient Temperature Range	0-55 °C
Ambient Humidity	90% max, non-condensing
Status LEDs	1 red, 1 green
Circuit Protection 3/2 Valve Spring to Open P	· Short circuit · Over-voltage · Under-voltage · Over-temp

Match motor current, voltage, command interface and operating profile. Specify wiring, enclosure and interlocks; this is an electrical accessory, not a fluid valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

DC Microstep Drive with Q Programming (ST5-Q-NN) V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

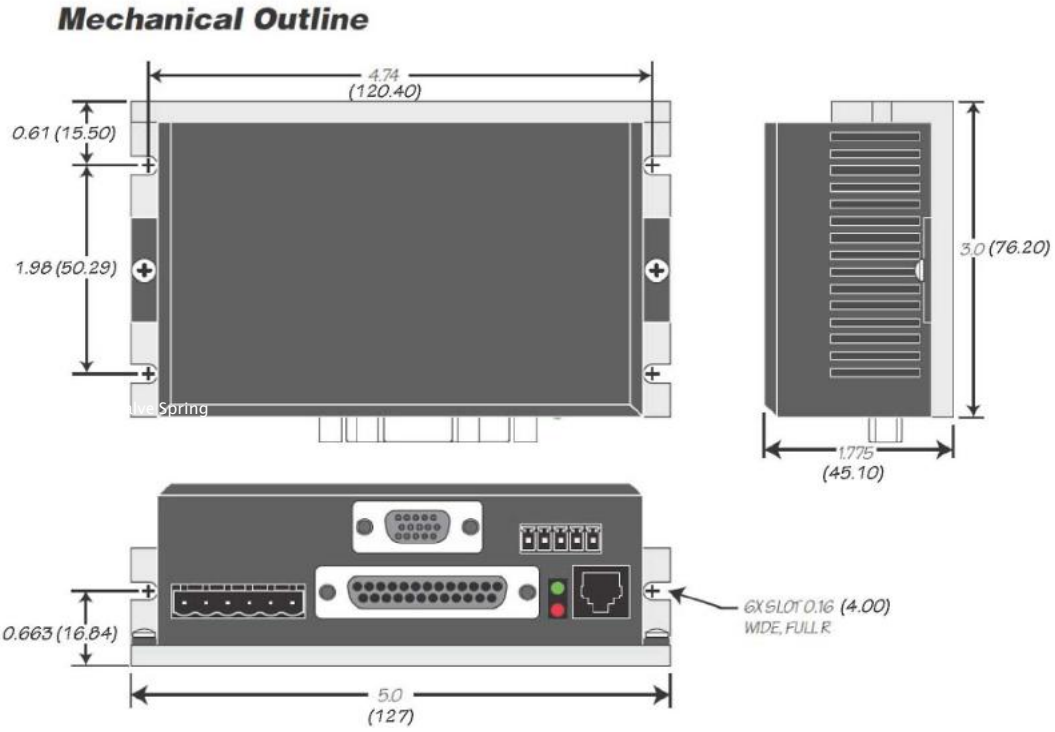
DC Microstep Drive with Q Programming (ST5-Q-NN)

Dimensions, connections & configuration | Manufacturer technical reference.

DOWNLOADS & SOFTWARE	
Manuals	ST5 10-Q QuickSetup 920-0007E.pdf Host Command Reference Rev I.pdf ST5-10-QSi Hardware Manual 920-0004F.pdf ModbusManual 920-0072.pdf
Datasheet	ST Datasheet 925-0007.pdf
2D Drawing	ST5 10 Dimensions.pdf ST T simple 3D.pdf
3D Drawing	ST5 10-Q Si C SIMPLE.igs
Speed-Torque Curves	ST speed-torque.pdf
Agency Approvals	ST-Q-Si-C-IP CE DOC.pdf
Application Notes	APPN0021 5V-Keepalive-Circuit.pdf APPN0019 Analog-positioning-using-Q-program.zip APPN0018_EZ-Series-Touchpanel-HMI.zip APPN0016_Simple-25-pin-mating-connections.pdf APPN0015_Make-a-serial-programming-cable.pdf
Software Downloads	SCL Utility ST Configurator™
Sample Code	scldemo.zip

All Downloads and Software can be downloaded from TWHC website

2D DRAWING



Match motor current, voltage, command interface and operating profile. Specify wiring, enclosure and interlocks; this is an electrical accessory, not a fluid valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

DC Microstep Drive with Q Programming (ST5-Q-NN) V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

DC Microstep Drive with Q Programming (ST5-Q-NN)

PRODUCTS IN THE SERIES *MODBUS PRODUCTS*

MODEL NUMBER	SUPPLY VOLTAGE	CONTROL MODES	OUTPUT CURRENT	COMMUNICATION PORTS
ST5-Q-NN	24-48 VDC	Step & Direction Velocity (Oscillator) Streaming Commands Analog Positioning Encoder Following Q Programming SiNet Hub Compatible Modbus	0.1-5.0 A/Phase	RS-232
ST5-Q-NN	24-48 VDC	Step & Direction Velocity (Oscillator) Streaming Commands Analog Positioning Encoder Following Q Programming Modbus	0.1-5.0 A/Phase	RS-232 RS-485

PRODUCTS IN THE SERIES *ST STEPPER DRIVES*

MODEL NUMBER	SUPPLY VOLTAGE	CONTROL MODES	OUTPUT CURRENT	COMMUNICATION PORTS
ST5-Q-NN	24-48 VDC	Step & Direction Velocity (Oscillator) Streaming Commands Analog Positioning Encoder Following Q Programming SiNet Hub Compatible Modbus	0.1-5.0 A/Phase	RS-232

*For Integrated encoder feedback consult TWHC

Match motor current, voltage, command interface and operating profile. Specify wiring, enclosure and interlocks; this is an electrical accessory, not a fluid valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

DC Microstep Drive with Q Programming (ST5-Q-NN) V.01.25.pdf, p. 4. Manufacturer labels and dimensions retained.

DC Microstep Drive with Q Programming (ST5-Q-NN)

Dimensions, connections & configuration | Manufacturer technical reference.

PART NUMBER	DESCRIPTION
561-002	Open Loop Flow Control Joystick Demand
561-003	Closed Loop Pressure Control Digital Demand
561-004	Closed Loop Pressure Control Analogue Demand 0-10v DC
561-005	Open Loop Flow Control Analogue Demand Uni-Directional 0-10v DC
561-006	Open Loop Flow Control Analogue Demand Bi-Directional 0-10v DC

Match motor current, voltage, command interface and operating profile. Specify wiring, enclosure and interlocks; this is an electrical accessory, not a fluid valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

DN16 Directional Valves

Selection data & performance | Manufacturer technical reference.

Pilot Operation

Actuation by water, oil or compressed air pilot is possible on all configurations. The water version is also available with an integral DN3 solenoid operated pilot valve. Please note: adequate flow must be available to ensure the full travel of the valve is made. The pilot piston can be sized to vary the pilot actuation pressure between 2 and 80 bar. The external pilot actuator is designed to operate at a pressure of 40 bar, although alternatives can be offered on request. For oil pilot, the actuator is supplied with a dual seal and air vent design to prevent cross contamination of the fluids.

Return Options

The valves shown are all spring return however, the valves can be offered with a detent function and pilot return if preferred.

SPECIFICATION	
Max Inlet Pressure	160 bar
Operating Pressure Range	0 to 100 bar
Flow Range	0-140 L/min
Flow Characteristics	CV Rate 5.0 Kv Rate 4.26
Feed Gallery Diameter	18mm
Porting	3/4" BSP (Parallel)
Construction Materials	A.I.S.I 316, Polymer & Martensitic Stainless steel
Pilot Operation	See Note
Pilot Media	Water, Air & Oil

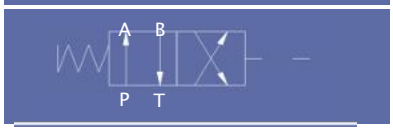
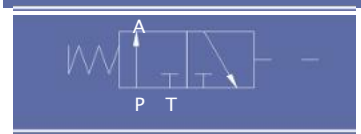
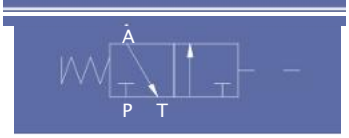


Functions:
4/2
3/2 N/O
3/2 N/C

The size of the pilot actuation can be modified to generate an actuation of different pilot pressure. For specific operating range contact TWHC. The DN16 is also available with an integral 3/2 solenoid operated pilot actuation valve. See 20*EESW for basic dimensions.

Electrical Actuation

Due to actuation forces and stroke length a direct solenoid operation is not possible with this size of valve. However, stepper motor control is under development.

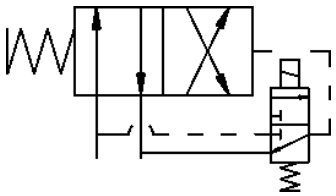
ORDERING CODES	Ext. Water Pilot	Oil Pilot	Air Pilot
	201EJSW	201ELSW	201EFSW
	208EJSW	208ELSW	208EFSW
	203EJSW	203ELSW	203EFSW

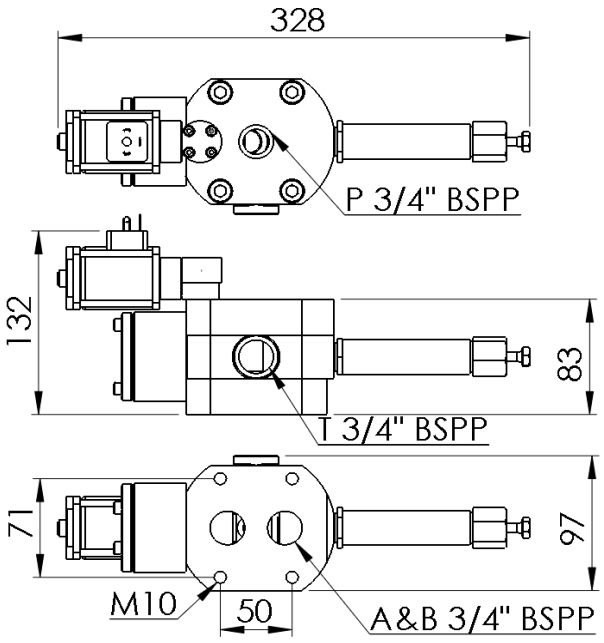
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DN16 Directional Valves

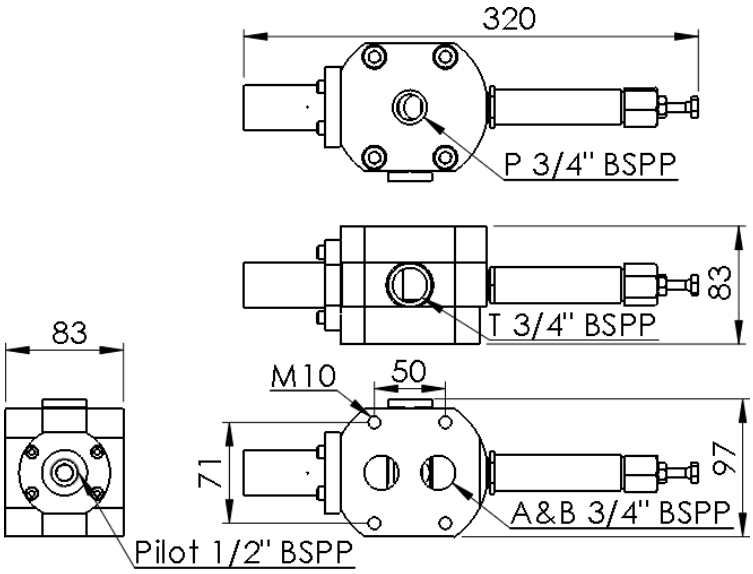
Dimensions, connections & configuration | Manufacturer technical reference.

201EDSW & 201EESW





201EJSW , 201ELSW & 201EFSW



Actuation housing size may vary with type of pilot and operating pressures.4/2 valve shown but also available in 3/2 with N/O or N/C

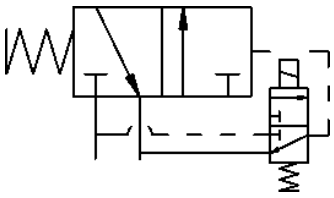
Size from actual flow and pressure drop. Confirm pilot supply, tank return, spool function, electrical supply and the selected manifold interface. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

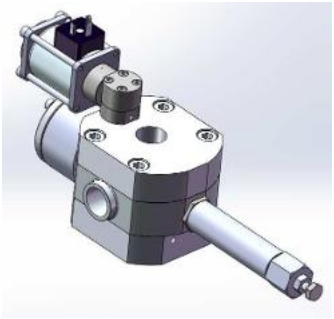
DN16 Directional Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

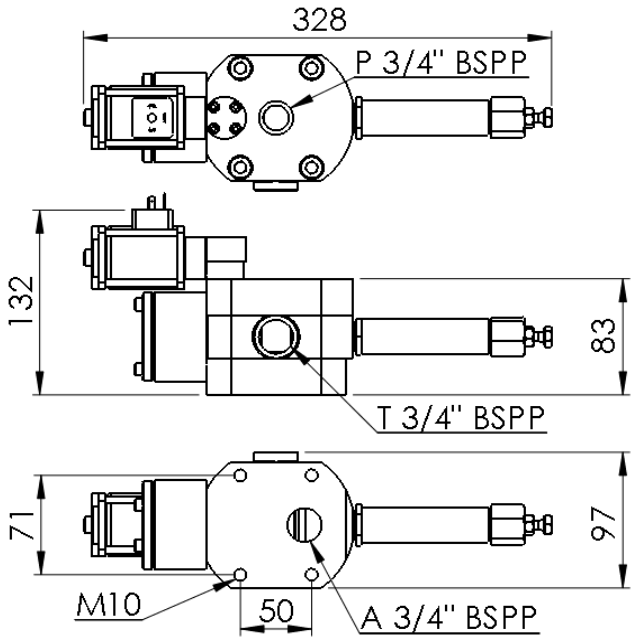
DN16 Directional Valves

Dimensions, connections & configuration | Manufacturer technical reference.

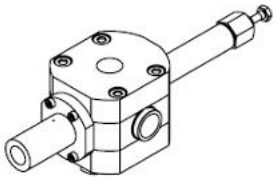
203EDSW & 203EESW

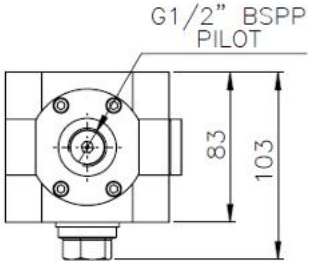


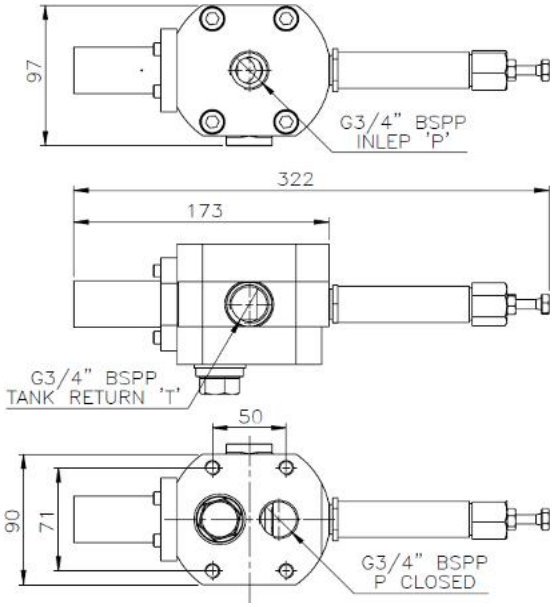




203EJSW , 203ELSW & 203EFSW







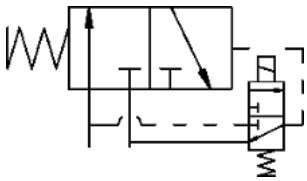
Size from actual flow and pressure drop. Confirm pilot supply, tank return, spool function, electrical supply and the selected manifold interface. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

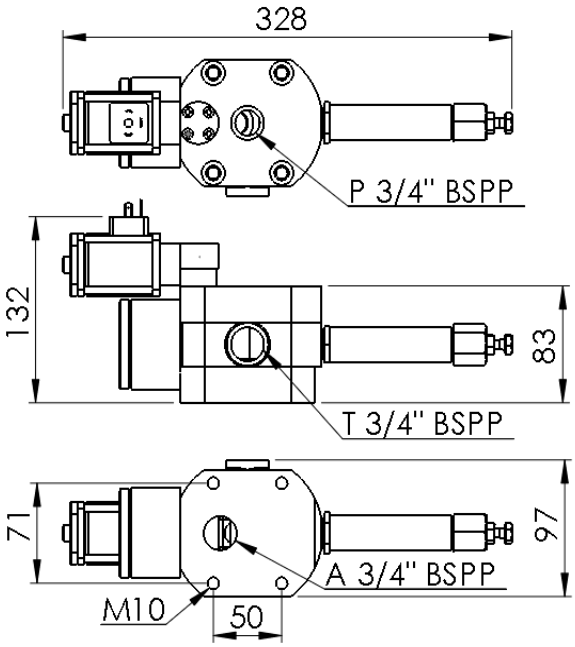
DN16 Directional Valves V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

DN16 Directional Valves

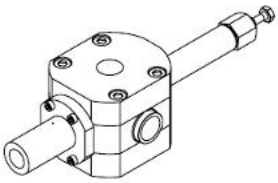
Dimensions, connections & configuration | Manufacturer technical reference.

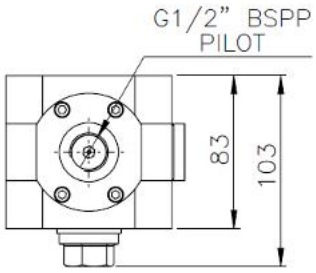
208EDSW & 208EESW

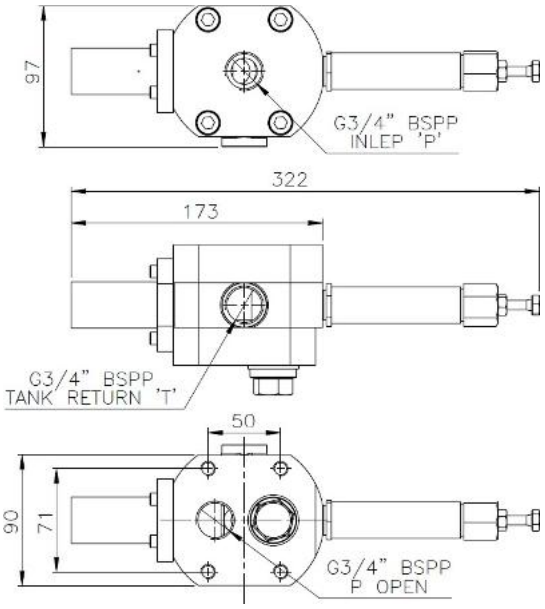




208EJSW , 208ELSW & 208EFSW







Size from actual flow and pressure drop. Confirm pilot supply, tank return, spool function, electrical supply and the selected manifold interface. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

DN16 Directional Valves V.01.25.pdf, p. 4. Manufacturer labels and dimensions retained.

Electronic Flow Control Valves

Selection data & performance | Manufacturer technical reference.

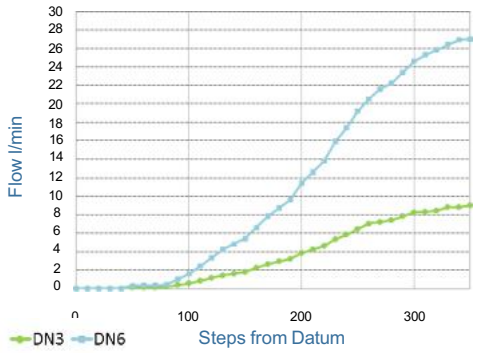
Design

The range of flow control valves are driven by a high accuracy stepper motor to ensure maximum precision for control by either open or closed loop means. The valves can be configured for manual/joystick input or to specific output speed/positional control. Closed loop feedback via output transducers is also possible with the use of an ST5 advanced micro-step programmable controller which is available with a 48V stable power supply if required. (Please see ST5 data sheet for further information). Additional output can also be offered for fail-safe configurations should such a function be necessary.
Software is available for purchase and downloadable via www.waterhydraulics.co.uk

SPECIFICATION	
Max Inlet Pressure	200 bar
Operating Pressure Range	0 to 160 bar
Flow Range	0-30 L/min
Feed Gallery Diameter	3mm & 6mm
Porting	BSP (Parallel) & Manifold
Construction Materials	316 Stainless Steel, Ceramic & Polymer
Motor Operation	48V DC
Electrical Connections	Flying lead or Plug
Current	125 mA



Flow Characteristics based on 100 bar supply pressure



Directional Flow Control Valve

Due to the very low viscosity of the fluid, the maximum orifice size (i.e. minimum pressure drop) for each application must be specified. The valve is therefore offered in either a 0.5-10L or a 5-30L port configuration to ensure maximum accuracy is gained across the full flow range. For fail safe function an additional N/C 2/2 valve is adopted in the P Line.

Pressure Control Valve

Available from 1 to 100l/min.
See pressure control valve data sheet for more information.

Additional on/off valves will need to be fitted if fail safes are required.

ORDERING CODES	STANDARD BUILD	MANIFOLD MOUNT	STANDARD BUILD	MANIFOLD MOUNT
	DN3 (1-10 L/min)		DN6 (3-30 L/min)	
 4/3 Valve Centre Position Closed P	223ARRW	2M23ARRW	223BRRW	2M23BRRW

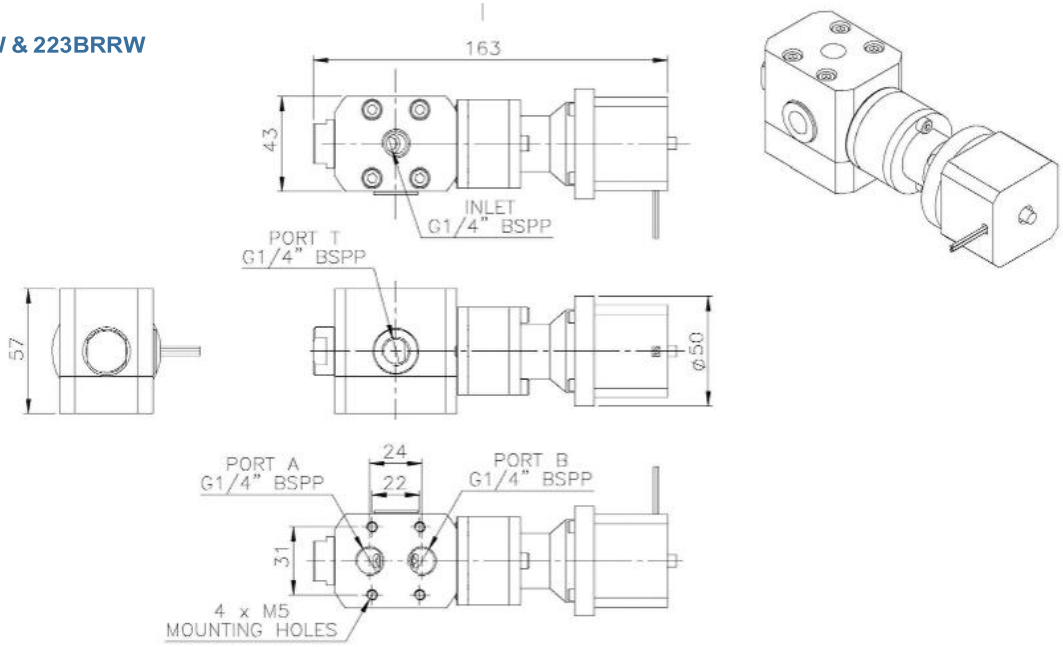
Specify commanded flow range, supply pressure and motor/control interface. Match the published 100 bar flow chart to its actual test condition. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Electronic Flow Control Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

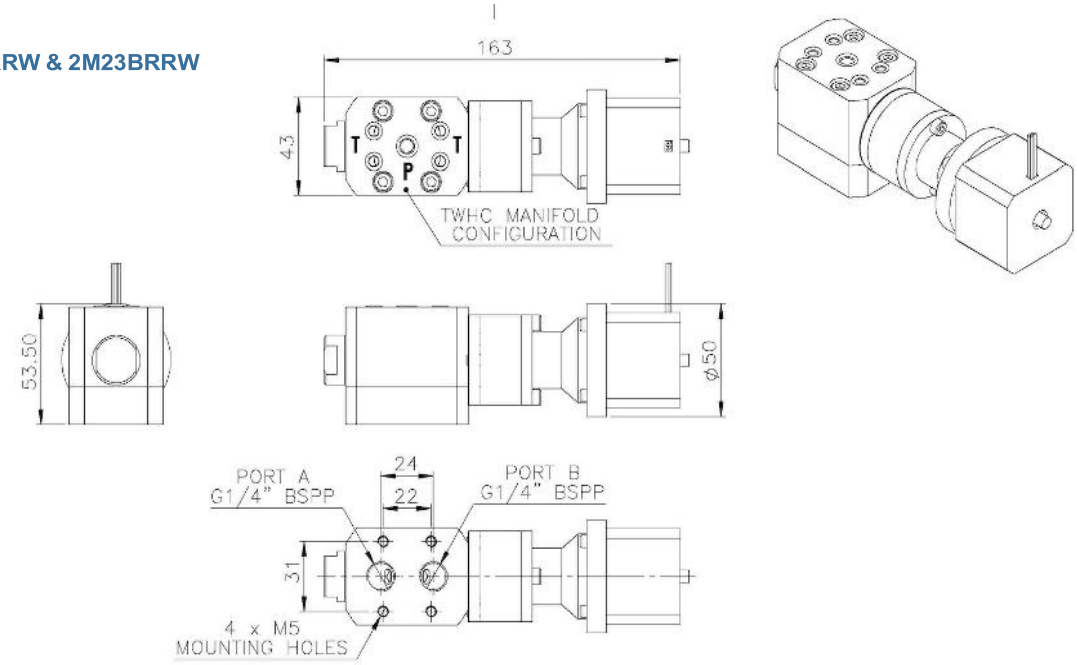
Electronic Flow Control Valves

Dimensions, connections & configuration | Manufacturer technical reference.

223ARRW & 223BRRW



2M23ARRW & 2M23BRRW



Specify commanded flow range, supply pressure and motor/control interface. Match the published 100 bar flow chart to its actual test condition. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Electronic Flow Control Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

Electronic Pressure Control Valve

Selection data & performance | Manufacturer technical reference.

Design

The range of proportional pressure control valves are driven by a high accuracy stepper motor to ensure maximum precision for control by either open or closed loop means. The valves can be configured for manual/joystick input or to specific output speed/ positional control. Closed loop feedback via output transducers is also possible with the use of an ST5 advanced micro-step programmable controller which is available with a 48V stable power supply if required. (Please see ST5 data sheet for further information). Additional output can also be offered for fail-safe configurations should such a function be necessary.
Software is available for purchase and downloadable via www.waterhydraulics.co.uk

SPECIFICATION		
	DN6	DN12
Max Inlet Pressure	200 bar	200 bar
Operating Pressure Range	0 to 160 bar	0 to 160 bar
Flow Range	0-30 L/min	5-100 L/min
Feed Gallery Diameter	6mm	12mm
Porting	BSP (Parallel) & Manifold	BSP (Parallel) & Manifold
Construction Materials	316 Stainless Steel& Polymer	316 Stainless Steel& Polymer
Motor Operation	48V DC	48V DC
Electrical Connections	Flying lead or Plug	Flying lead or Plug



Pressure Control Valve

This valve is available with open or closed loop control in conjunction with a 0.5V pressure transducer input. The ST5 programmer comes with an on-board PLC complete with programme, to offer stand-alone, turn-key operation.

ORDERING CODES	STANDARD BUILD	MANIFOLD MOUNT	STANDARD BUILD
	DN6 (1-20 L/min)		DN12 (5-100 L/min)
 Electronic Pressure Control Valve	225BRRW	2M25BRRW	225DRRW

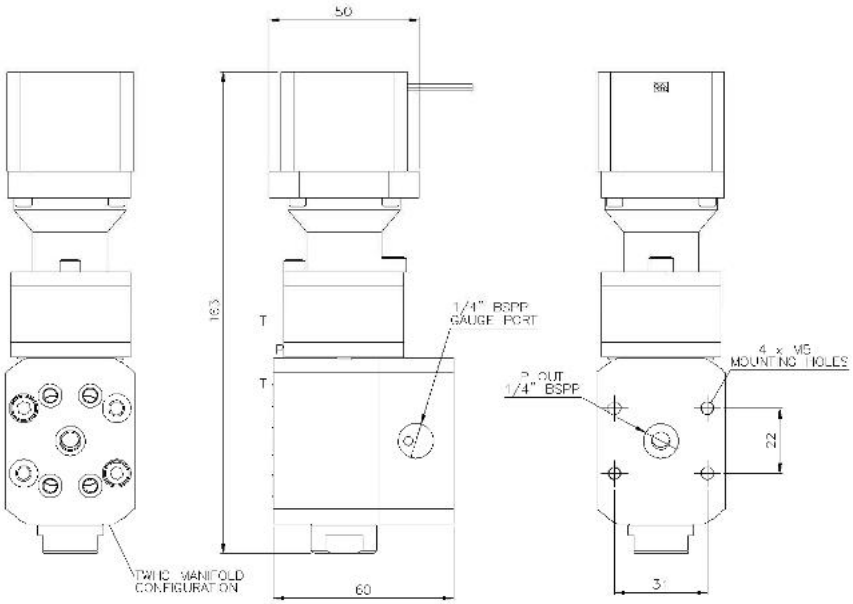
Define pressure range, ramp and cyclic duty. Check feedback, driver interface and protection; the source typical cyclic response is limited to 1 Hz. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Electronic Pressure Control Valve V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

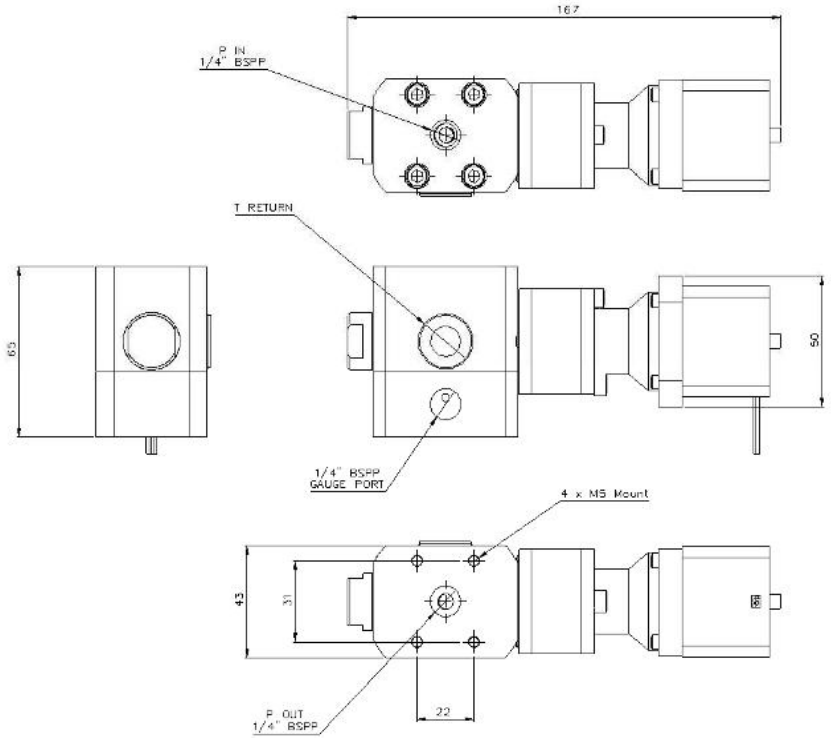
Electronic Pressure Control Valve

Dimensions, connections & configuration | Manufacturer technical reference.

2M25BRRW



225BRRW



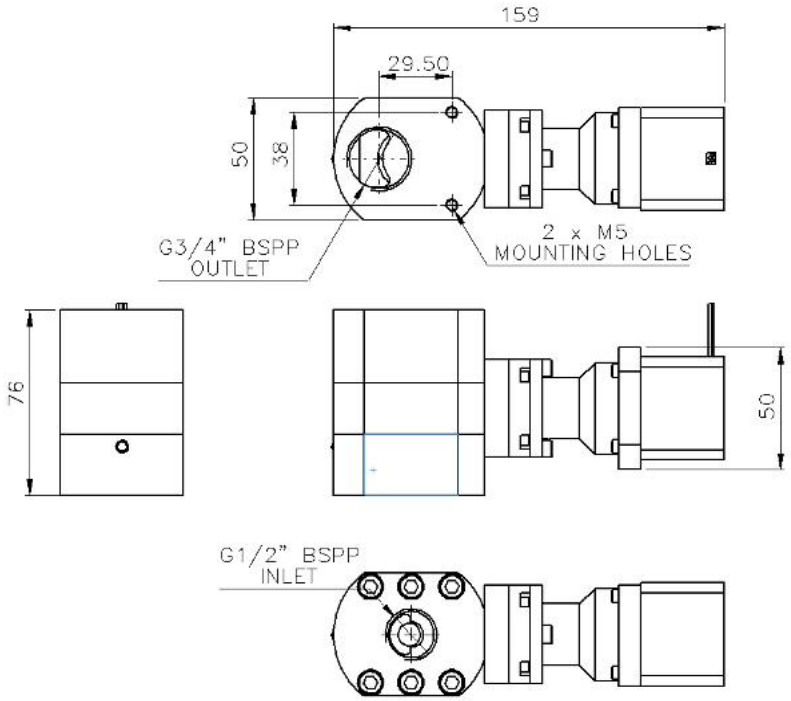
Define pressure range, ramp and cyclic duty. Check feedback, driver interface and protection; the source typical cyclic response is limited to 1 Hz. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Electronic Pressure Control Valve V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

Electronic Pressure Control Valve

Dimensions, connections & configuration | Manufacturer technical reference.

225DRRW



Define pressure range, ramp and cyclic duty. Check feedback, driver interface and protection; the source typical cyclic response is limited to 1 Hz. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

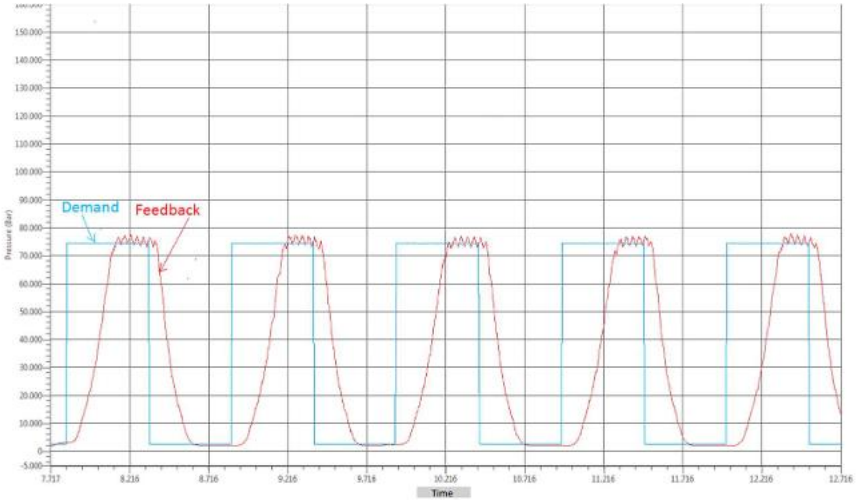
Electronic Pressure Control Valve V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

Electronic Pressure Control Valve

Dimensions, connections & configuration | Manufacturer technical reference.

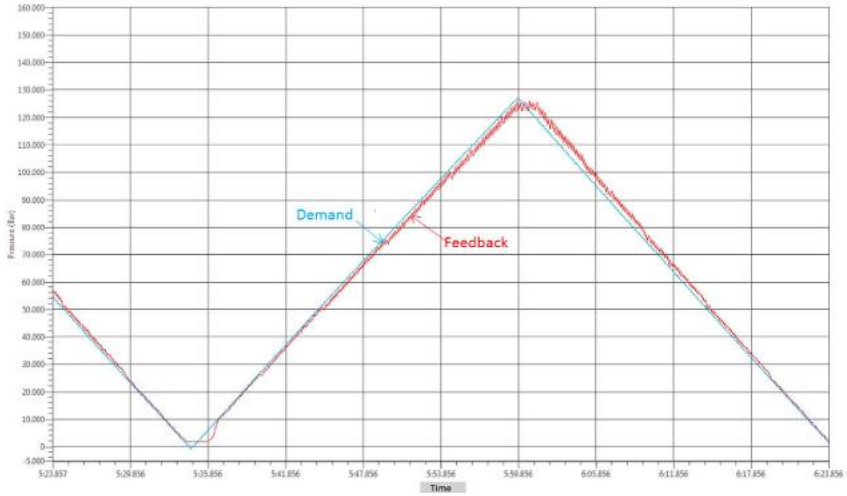
Cyclic Pressure Test

Typical performance graph for high frequency pulsation system with a maximum cycle of 1Hz.



Controlled Ramping Pressure

Between 1-160 bar



The accuracy of both applications can be trimmed by honing the pseudo PID control parameters, size of test piece (i.e. pressurised volume), degree of air entrapment and test piece expansive volume. This will all affect the accuracy however, simple programme parameters can be adjusted to meet the demand within the precision of the transducer.
Filtering of the transducer can also offer an improved performance.

Define pressure range, ramp and cyclic duty. Check feedback, driver interface and protection; the source typical cyclic response is limited to 1 Hz. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Electronic Pressure Control Valve V.01.25.pdf, p. 4. Manufacturer labels and dimensions retained.

Flow Control Valve

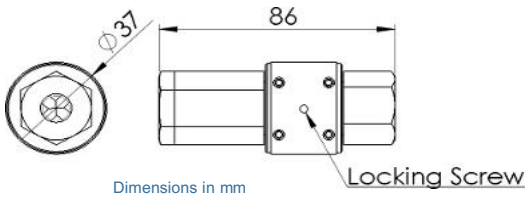
Selection data & performance | Manufacturer technical reference.

Manufactured from 316 stainless steel and special engineering polymer selected for their corrosion resistance and resilience to flow erosion. Designed for operation on tap water to EU directive 98/83/EC without additive.

SPECIFICATION		
	DN3	DN6
Operating Pressure Range	0.1 to 200 bar	0.1 to 200 bar
Maximum Cv Value in flow control direction	0.02	0.62
Feed Gallery Diameter	3mm	6mm
Porting	1/4" BSP (Parallel)	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer	A.I.S.I 316/Polymer
Number of turns Fully open to fully closed	4	4
Media	Water	Water



Dimensions

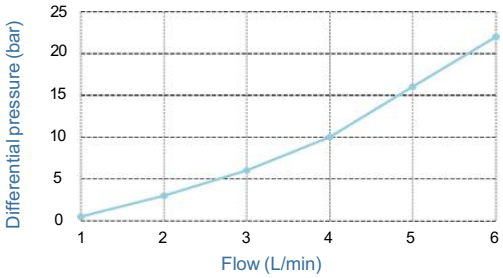


Dimensions in mm

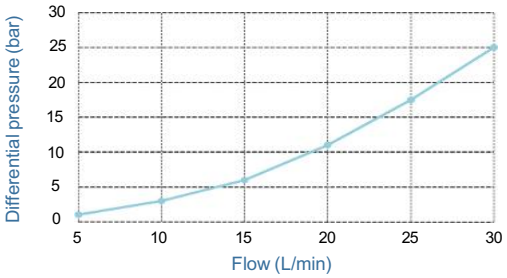
Rotation of the control wheel opens the flow to a series of helical ports resulting in a very precise control throughout its range. Using materials with a high resistance to flow erosion and a design that ensures the aggressive high pressure jets do not affect the long term setting of the valve.

The valve is supplied with an externally adjusted thumb wheel for operator control without the need for tools. Also fitted with a tamper proof locking screw; when locked, the wheel can be rotated but the setting will not be affected.

Free Flow Characteristics DN3



Free Flow Characteristics DN6



ORDERING CODES	DN3	DN6
 Flow Control Valve	204AXXW	204BXXW

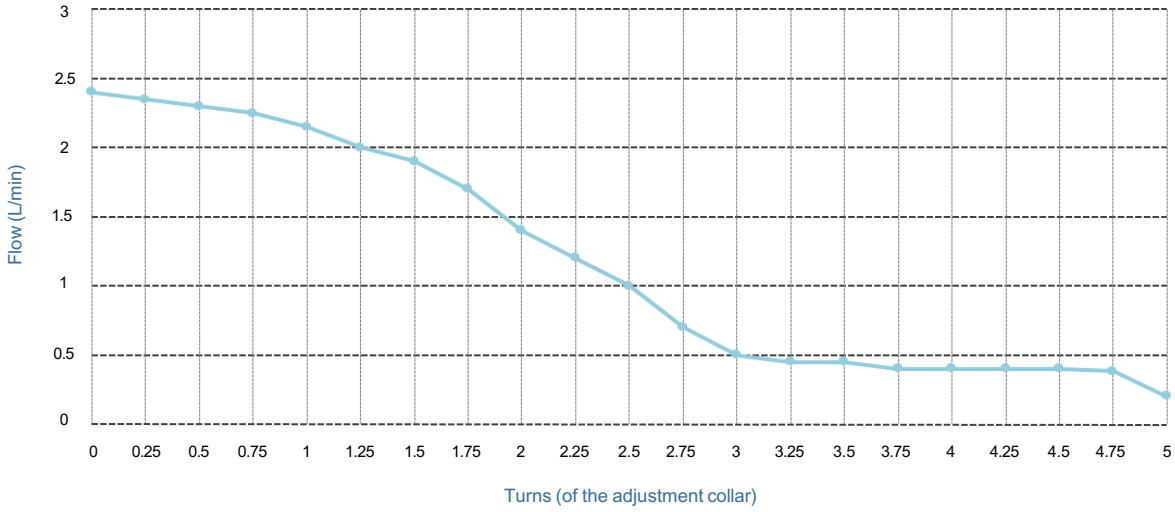
Identify the metered direction and free-return direction. Flow depends on differential pressure; adjustment-turn curves apply to the stated test pressure. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Flow Control Valve V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

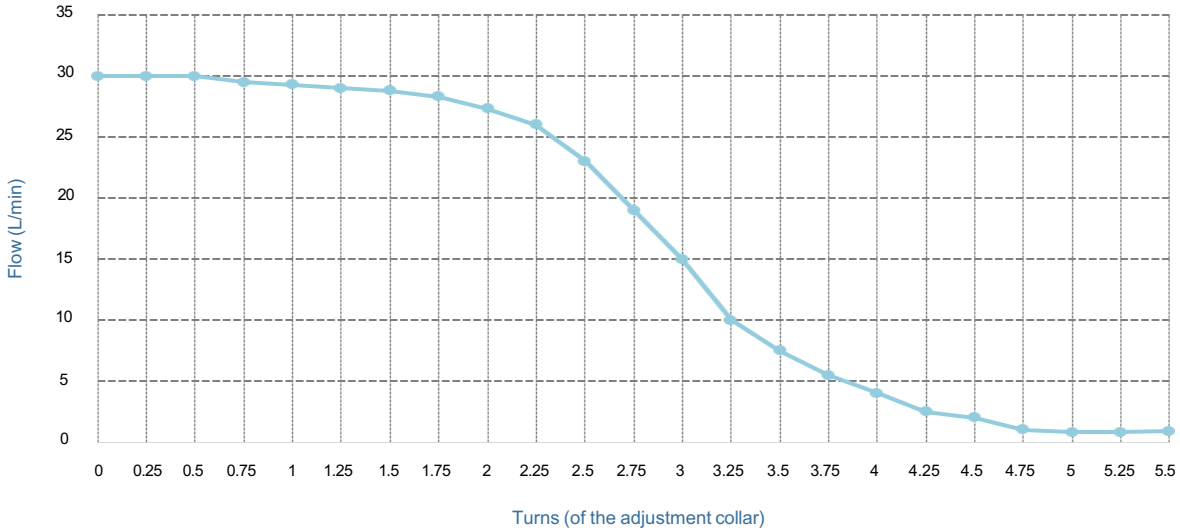
Flow Control Valve

Dimensions, connections & configuration | Manufacturer technical reference.

204AXXW Restricted Flow output in relation to turns of the collar 160 bar inlet pressure



204BXXW Restricted Flow output in relation to turns of the collar 160 bar inlet pressure



Identify the metered direction and free-return direction. Flow depends on differential pressure; adjustment-turn curves apply to the stated test pressure. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Flow Control Valve V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

Hose Burst Valve

Selection data & performance | Manufacturer technical reference.

Manufactured from 316 stainless steel and special engineering polymer selected for their corrosion resistance and resilience to flow erosion. Designed for operation on tap water to EU directive 98/83/EC without additive.

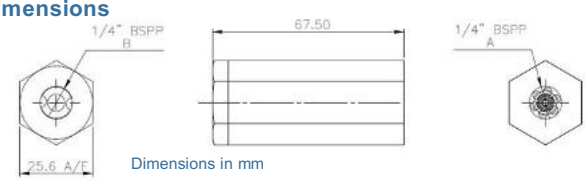
SPECIFICATION		
	DN3	DN6
Operating Pressure Range	0.1 to 200 bar	0.1 to 200 bar
Reverse Flow Adjustment L/min	1 to 5	5-10
Max Flow rate L/min	30	50
Maximum Cv Value in flow control direction	0.02	0.62
Feed Gallery Diameter	6mm	6mm
Porting	1/4" BSP (Parallel)	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer	A.I.S.I 316/Polymer
Media	Water	Water



The hose burst valves allows free flow in both directions under normal operating conditions, however in the event of a hose burst the surge in flow from 'B' to 'A' the valve will immediately stop the flow locking pressure in 'B' . The valve can be reset by reversing the flow when the hose has been replaced. Reaction flow rate should be set to 1.5 x maximum system flow rate.

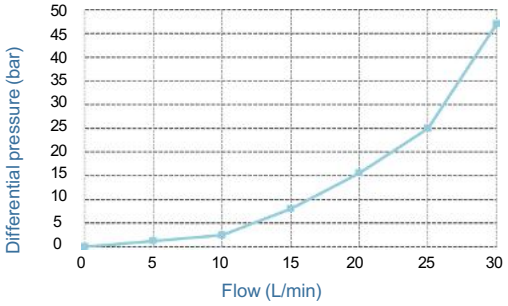
The valve is supplied with an internally adjusted spring. This can be set by TWHC to the flow required.

Dimensions

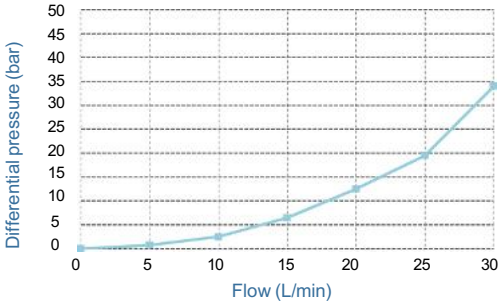


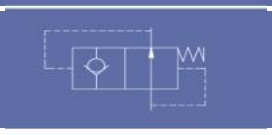
Dimensions in mm

Free Flow Characteristics DN3



Free Flow Characteristics DN6



ORDERING CODES	DN3	DN6
 Hose Burst Valve	591-001	591-007

Match trip flow to normal motion, transients and the protected load. Verify reset, leakage and response within the complete machine circuit. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

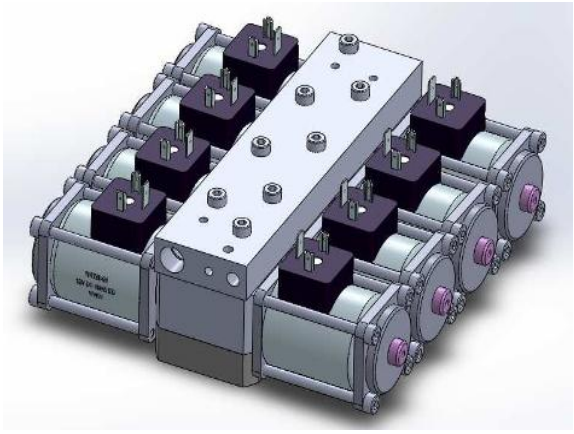
Hose Burst Valve V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

Manifolds

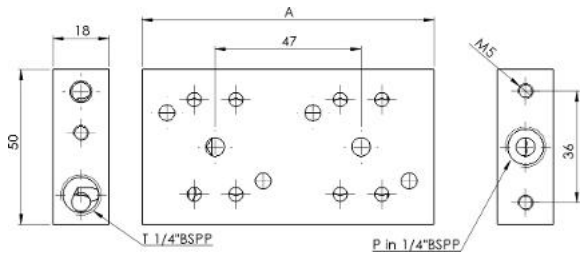
Selection data & performance | Manufacturer technical reference.

Available in 1 to 4 stations including with or without additional relief valve porting and p[illegible] will accept the 3/2, 4/2 and 4/3 valve configurations in all actuation formations such as solenoid, pilot or manual.
Maximum flow rate is 30 l/min without integral relief and 22l/min if a 2M07BXSW is incorporated.

NO. OF STATIONS	1	2	3	4
Dimension A	47	94	141	188



Dimensions

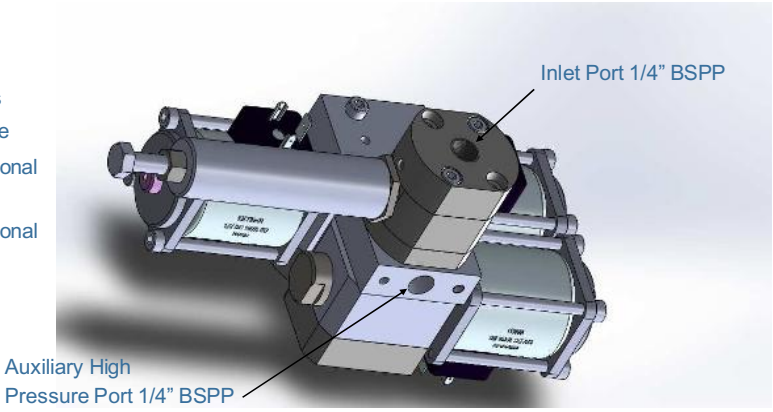


Assembly Layout Example

Consisting of:

- 6manir2: 2 station manifold including relief ports
- 2M07BXSW: DN6 manifold mounted relief valv e
- 2M03BBSW: DN6 manifold mounted 3/2 directi onal control valve.
- 2M06BBSW: DN6 manifold mounted 4/3 directi onal control valve.

The relief valve has 2 through bolts which fasten through the manifold into the 3/2 valve opposite.



ORDER ING CODE	1 STATION	2 STATION	3 STATION	4 STATION
STAND ALONE	6MANI1	6MANI2	6MANI3	6MANI4
INCLUDING RELIEF PORTS	6MANIR1	6MANIR2	6MANIR3	6MANIR4

Provide the full circuit, station functions, maximum pressure and flows. Confirm port/thread standards, mounting, proof test and external connection access. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Manifolds V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

PO Check Valve (DN12)

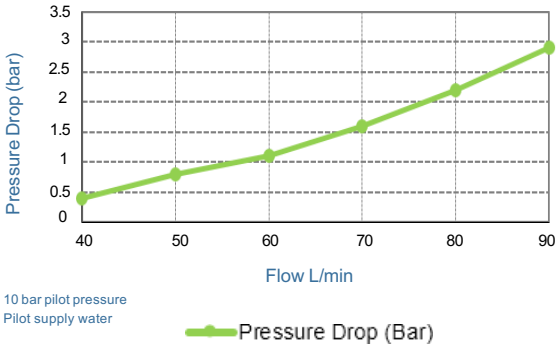
Selection data & performance | Manufacturer technical reference.

Supplied in all 316 stainless steel construction, the DN12 PO Check can be actuated by multiple fluids; the valves are double sealed to ensure cross contamination does not occur. The PO Check offers bubble tight 100% seal operation for load or pressure holding applications and is available as stand alone or manifold mounted. The manifold block can be configured to offer double port checked functions.

SPECIFICATION	
Max Inlet Pressure	160 bar
Operating Pressure Range	0 to 160 bar
Flow Range	0-60 L/min
Flow Characteristics	CV Rate 3.9 Kv Rate 0.75
Feed Gallery Diameter	12mm
Porting	1/2" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer



Flow Characteristics



The standard valve has a 2.3:1 ratio allowing the piston pressure to be overcome by a much lower actuation pressure.

This will allow the actuation of the valve to be controlled with min-imum input pressure and over coming amplification pressures due to cylinder differential areas. As a special request, this ratio can be modified to suit your system requirements.

ORDERING CODES	STANDARD BUILD	MANIFOLD MOUNT
 Single PO Check Valve 2.3:1 Ratio	211DJSW	2M11DJSW

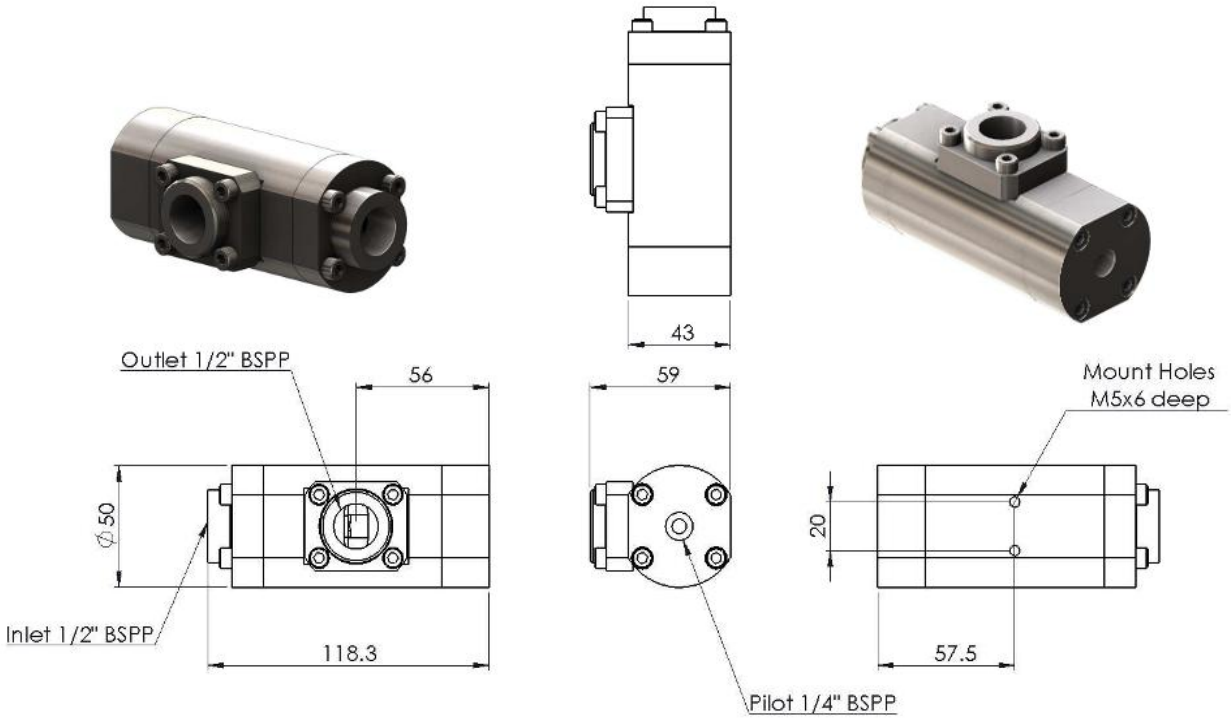
Verify pilot pressure, pilot ratio, trapped-load release and permissible leakage. Account for pressure intensification and the required load-holding function. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

PO Check Valve (DN12) V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

PO Check Valve (DN12)

Dimensions, connections & configuration | Manufacturer technical reference.

211DJSW



Verify pilot pressure, pilot ratio, trapped-load release and permissible leakage. Account for pressure intensification and the required load-holding function. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

PO Check Valve (DN12) V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

PO Check Valve (DN6)

Selection data & performance | Manufacturer technical reference.

SPECIFICATION	
Max Inlet Pressure	200 Bar
Operating Pressure Range	0 to 160 Bar
Flow Range	0-30 L/min
Flow Characteristics 3:1 9:1	CV Rate 0.62 Kv Rate 0.75 Cv Rate 0.41 Kv Rate 0.50
Feed Gallery Diameter 3:1 9:1	6mm 4mm
Porting	1/4" BSP (Parallel)
Construction Materials	A.I.S.I 316/Polymer
Media	Water
Cracking Pressure Free Flow	6 bar standard 1 bar available on request

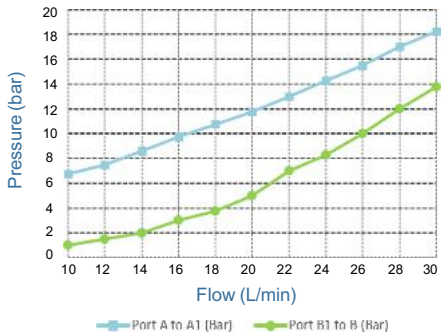
Available as an individual valve or as a bolt on option to the 4/3 and 4/2 directional control valve, the PO Check offers bubble tight 100% seal operation for load or pressure holding applications.

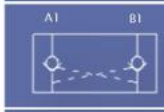
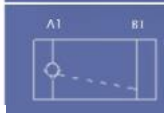
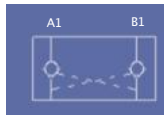
To accommodate for a wide range of pilot operating pressures, the valves are offered in 3:1 and 9:1 pilot to seal area ratios. This will allow the actuation of the valve to be controlled with minimum input and prevent operation from residual pressures.



They are offered as either single or double port checked. The valve can be used for locking a water cylinder in a stationary position. Actuation can only occur when the required pilot pressure is available. A port adaptor block is also available for remote use, e.g. cylinder mounted.

Flow Characteristics



ORDERING CODES		STANDARD BUILD	MANIFOLD MOUNT
	Double PO Check Valve 3:1 Ratio	213BJSW	2M13BJSW
	Single PO Check Valve 3:1 Ratio	211BJSW	2M11BJSW
	Double PO Check Valve 9:1 Ratio	222BJSW	2M22BJSW
	Single PO Check Valve 9:1 Ratio	219BJSW	2M19BJSW

V01

Verify pilot pressure, load pressure and the release sequence. Check leakage and use the chosen valve as part of the complete holding circuit. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

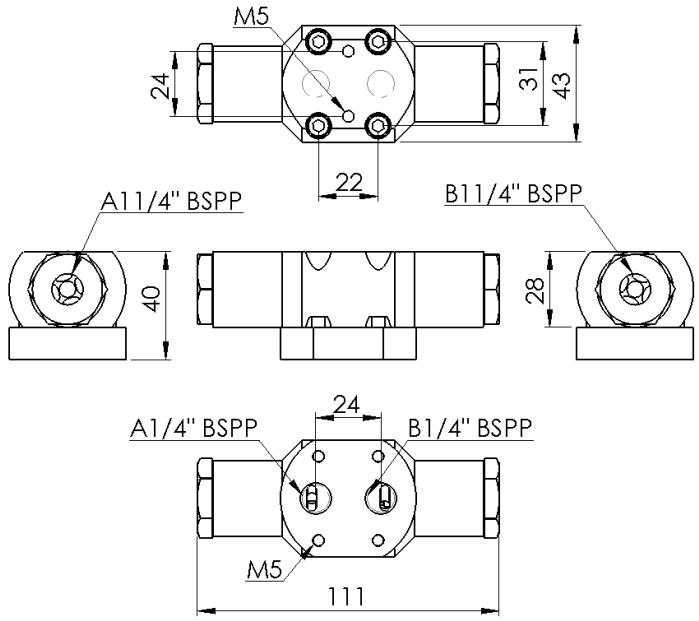
PO Check Valve (DN6) V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

PO Check Valve (DN6)

Dimensions, connections & configuration | Manufacturer technical reference.

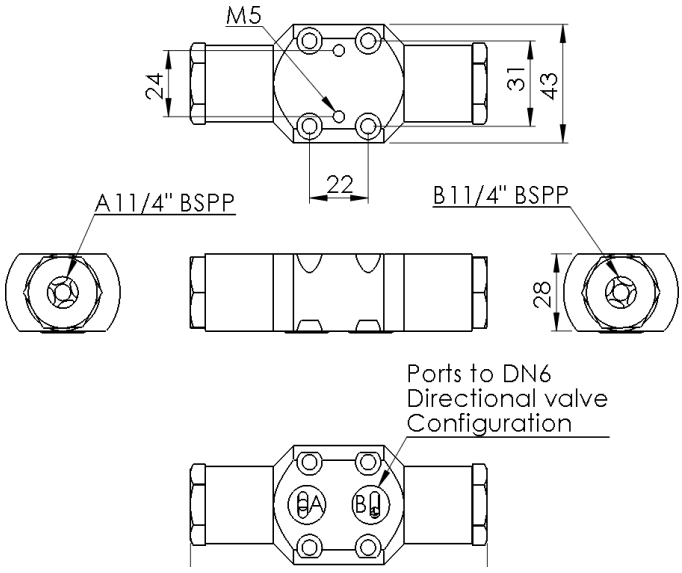
213BJSW & 222BJSW





2M13BJSW & 2M22BJSW





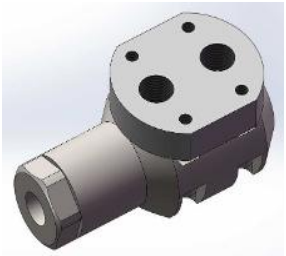
Verify pilot pressure, load pressure and the release sequence. Check leakage and use the chosen valve as part of the complete holding circuit. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

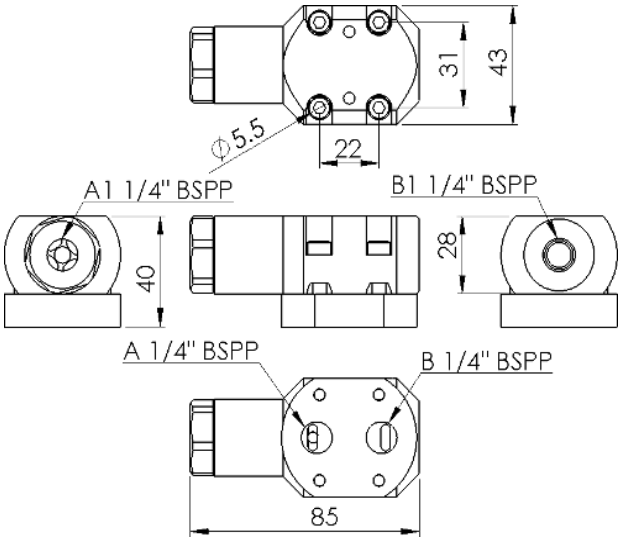
PO Check Valve (DN6) V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

PO Check Valve (DN6)

Dimensions, connections & configuration | Manufacturer technical reference.

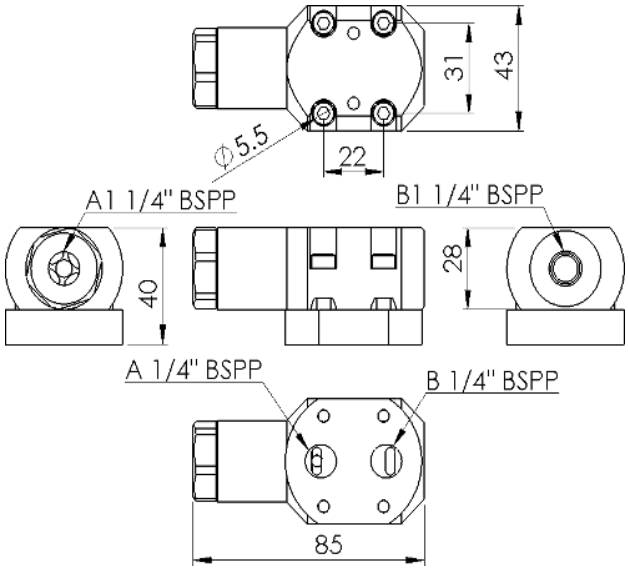
211BJSW & 219BJSW





2M11BJSW & 2M19BJSW





Verify pilot pressure, load pressure and the release sequence. Check leakage and use the chosen valve as part of the complete holding circuit. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

PO Check Valve (DN6) V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

Pressure Compensating Flow Control Valve

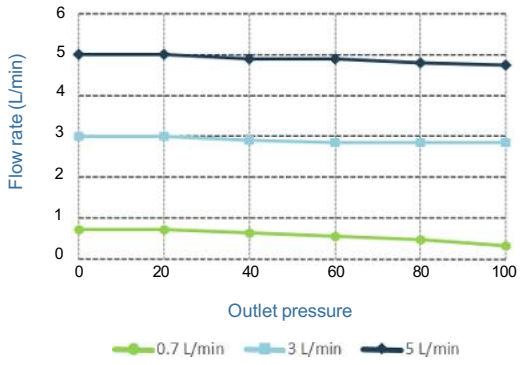
Selection data & performance | Manufacturer technical reference.

The pressure compensated flow control valve is designed to control the flow rate of tap drinking water in a hydraulic system. Fully adjustable between the specified ranges, the output flow can be regulated and maintained at a constant level irrespective of the output working pressure. The constant rate can be adjusted by allen key then loaded to prevent none desired alteration. The reverse flow gives a slightly restrictive orifice but vertically free for one way control. See graph for characteristics.

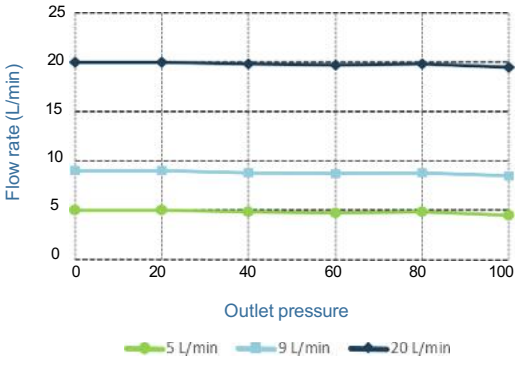
SPECIFICATION		
	DN3	DN6
Operating Pressure Range	5-160 bar	5-160 bar
Maximum Cv Value in flow control direction	0.2	0.7
Feed Gallery Diameter	3mm	6mm
Porting	1/4" BSPP	1/4" BSPP
Construction Materials	316 Stainless Steel Polymer	316 Stainless Steel Polymer
Number of turns Fully open to fully closed	5	6
Media	Tap Water	Tap Water
Flow Range	0.5—5 L/min	3—30 L/min



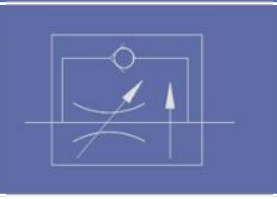
Flow Characteristics DN3



Flow Characteristics DN6



Inlet pressure/ Relief setting 130 bar
Minimum operating pressure 5 bar

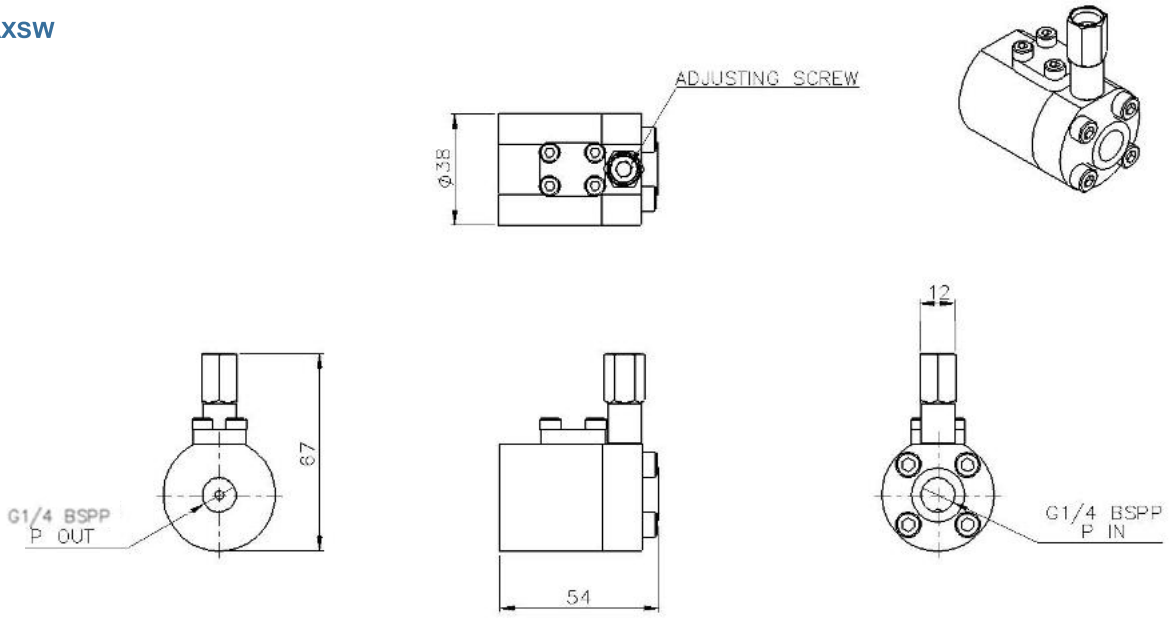
ORDERING CODES	DN3	DN6
	216AXSW	216BXSW

Select regulation range and allow the pressure margin needed to compensate. Check minimum operating pressure and heat loss across the valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

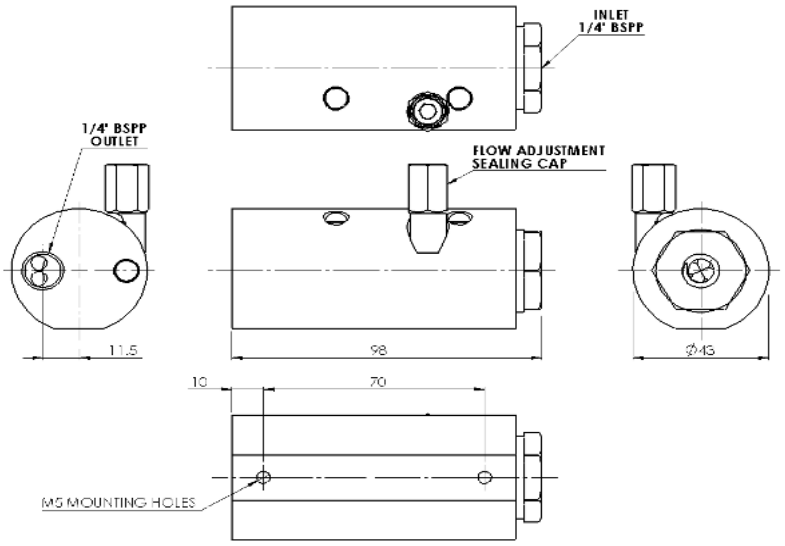
Pressure Compensating Flow Control Valve

Dimensions, connections & configuration | Manufacturer technical reference.

216AXSW



216BXSW



Select regulation range and allow the pressure margin needed to compensate. Check minimum operating pressure and heat loss across the valve. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Pressure Reducing Valves

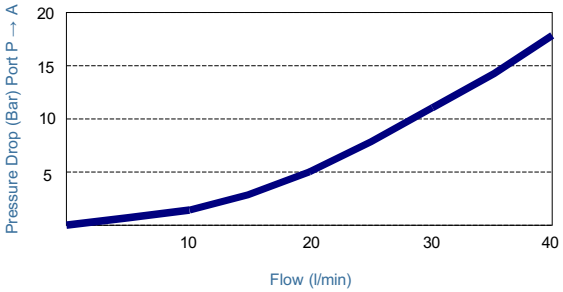
Selection data & performance | Manufacturer technical reference.

A patented range of valves designed to control pressure in a hydraulic system using tap water to EC directive 98/83/EC. Designed specifically for fluids with low viscosity, the valve offers accurate control with high resistance to flow erosion experienced in pressure control applications.

SPECIFICATION	
Max Inlet Pressure	180 Bar
P Out Setting Range	1 10 to 30 Bar 2 30 to 100 Bar 3 70 to 160 Bar
Maximum Flow	0-30 L/min
Feed Gallery Diameter	6mm
Porting	1/4" BSP (Parallel)
Construction Materials	316 Stainless Steel & Polymer
Media	Water



Flow Characteristics



Function

Designed to protect equipment from high system pressure in multi function circuits. An example of use is to limit the force of a cylinder when higher system pressure is needed for other functions. The valve is designed to limit the pressure in the P out port even if there is zero flow demand in this line. This function ensures cylinder force control even under stall or reaching an end stop condition. The valve restricts the inlet when nearing the control pressure setting and with the same action, relieves the service line to tank ensuring only minimal energy is lost from the system.

The pressure can be reduced to a maximum of 70% of the inlet pressure in one stage, e.g. 100 bar system pressure, minimum reduced pressure 30 bar.

Reverse Flow

The valve does not offer a restriction with reverse flow.

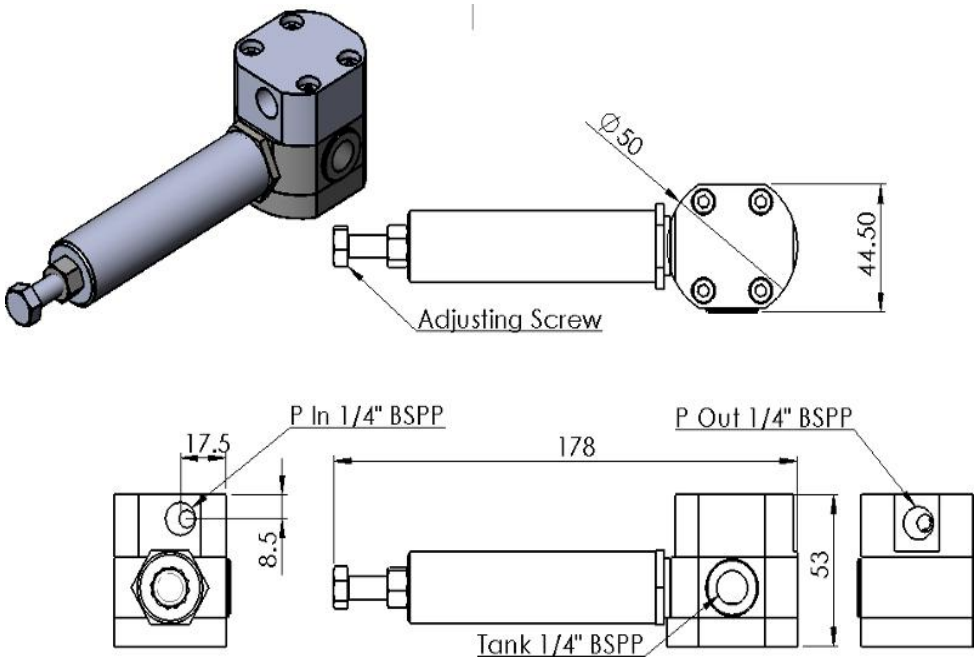
ORDERING CODE	DN6
 Reducing Valve 10 to 30 bar	109BXSW
 Reducing Valve 30-100 bar	209BXSW
 Reducing Valve 70-160 bar	309BXSW

Define downstream set pressure, supply variation and flow. Check minimum regulating differential, drain/backpressure effects and relieving requirements. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Pressure Reducing Valves

Dimensions, connections & configuration | Manufacturer technical reference.

109BXSW, 209BXSW & 209BXSW



Define downstream set pressure, supply variation and flow. Check minimum regulating differential, drain/backpressure effects and relieving requirements. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Relief Valves

Selection data & performance | Manufacturer technical reference.

A patented range of valves which can control the pressure of corrosive fluids in hydraulic systems. Designed specifically for fluids with low viscosity, the valve offers accurate control with high resistance to flow erosion experienced in pressure control applications. The valve fully conforms to the Pressure Equipment Directive and can include internal burst disc assemblies for safety applications.

SPECIFICATION	
Max Inlet Pressure	180 Bar
Operating Pressure Range	1 10 to 30 Bar 2 70 to 160 Bar 3 >160 bar 4 25 to 100 Bar
Flow Range - DN3 - DN6 - DN12 - DN16	0-10 L/min 0-30 L/min 0-120 L/min 0-450 L/min
Construction Materials	316 Stainless Steel Polymer or Ceramic
Media	Water



Seals

Standard valves are supplied with Nitrile seals however, alternatives are available on request.

Noise

The valve offers very low noise output even when relieving at high system pressures. The usual high pitch “squeal” associated with water relief valves is not generated with this range of valves.

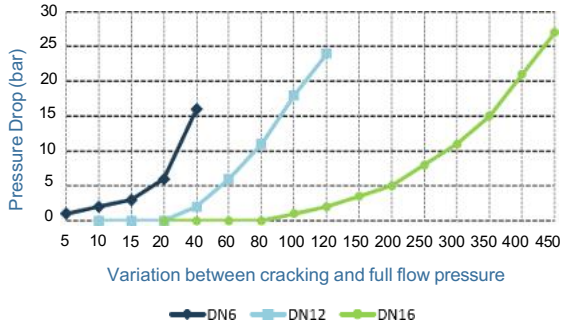
Relief Setting

The operating pressure range must be specified on all orders. Not only the spring force varies with pressure but also the actuation piston is sized to give both accuracy and precision in low, medium and high pressure valves.

Drip rate

The valves do not offer 100% seal below set pressure. A drip rate when operating at 75% of its set pressure will result.

Flow Characteristics



Variation between cracking and full flow pressure

ORDERING CODE	DN6	DN6 MANIFOLD	DN12	DN16
10-30 bar	107BXSW	1M07BXSW	107DXSW	107EXSW
25-100 bar	407BXSW	4M07BXSW	407DXSW	407EXSW
70-160 bar	207BXSW	2M07BXSW	207DXSW	207EXSW
> 160 bar	307BXSW	3M07BXSW	307DXSW	307EXSW

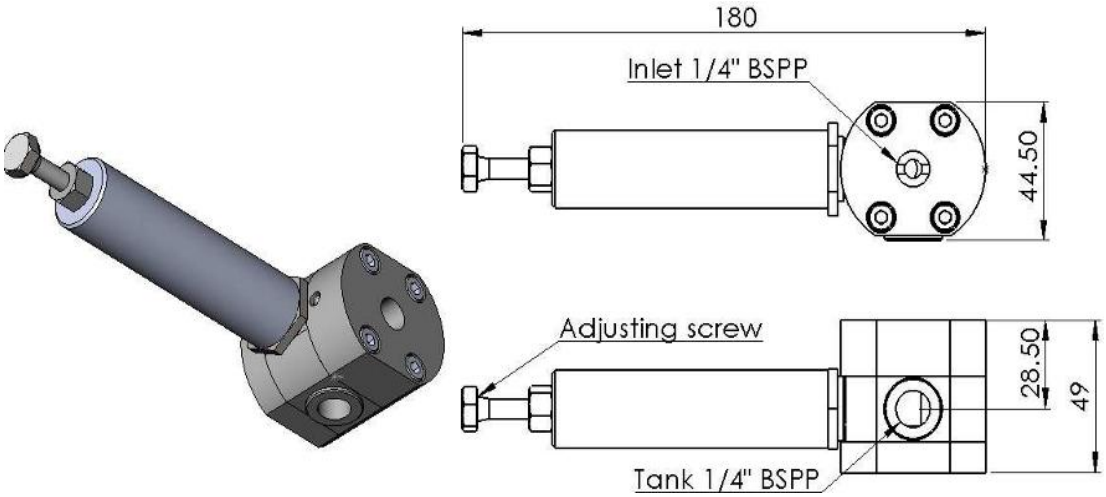
V01.25 part of any contract, whether expressed or implied. The publishers reserve the right to amend specification at their discretion.

Set range and relieving capacity must suit the protected circuit. DN12 capacity: 0-120 L/min. Chart flow ticks have unequal numeric steps at equal spacing; do not interpolate by horizontal distance. It is not a zero-leakage isolation valve; verify return pressure and allowable override. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Relief Valves

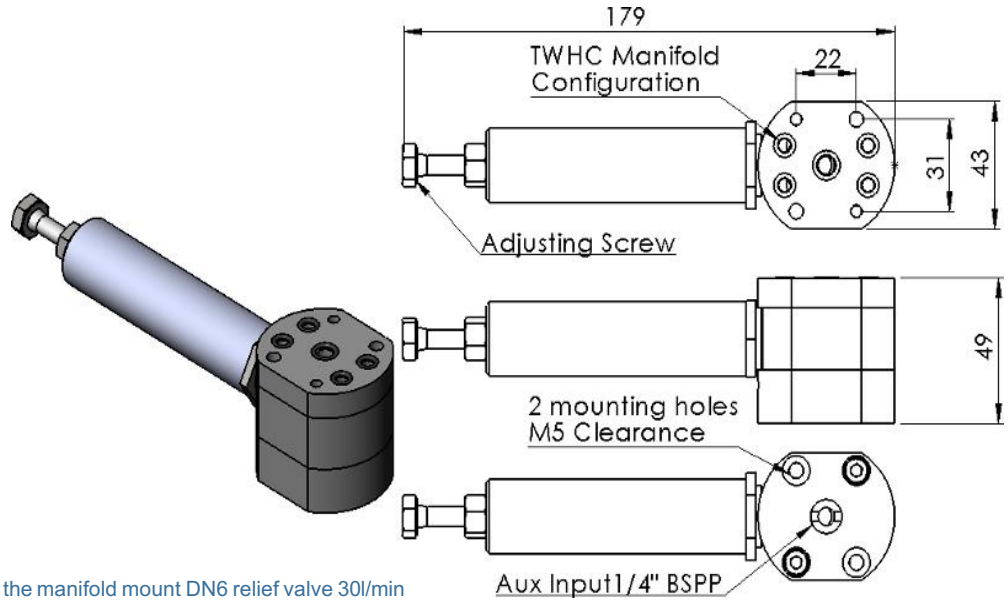
Dimensions, connections & configuration | Manufacturer technical reference.

207BXSW



Maximum flow on the stand alone DN6 relief valve 30l/min

2M07BXSW



Maximum flow on the manifold mount DN6 relief valve 30l/min

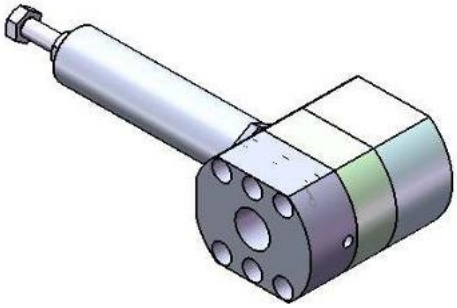
Set range and relieving capacity must suit the protected circuit. DN12 capacity: 0-120 L/min. Chart flow ticks have unequal numeric steps at equal spacing; do not interpolate by horizontal distance. It is not a zero-leakage isolation valve; verify return pressure and allowable override. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

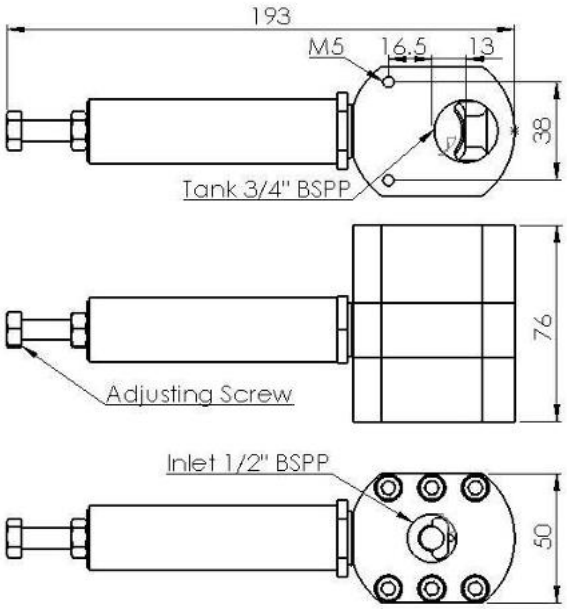
Relief Valves V.01.25.pdf, p. 2. Manufacturer labels and dimensions retained.

Relief Valves

Dimensions, connections & configuration | Manufacturer technical reference.

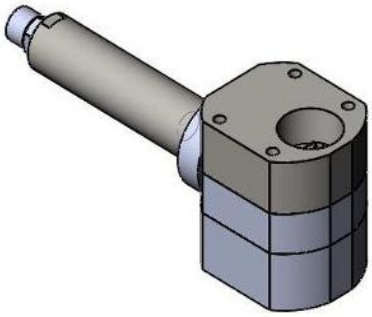
207DXSW

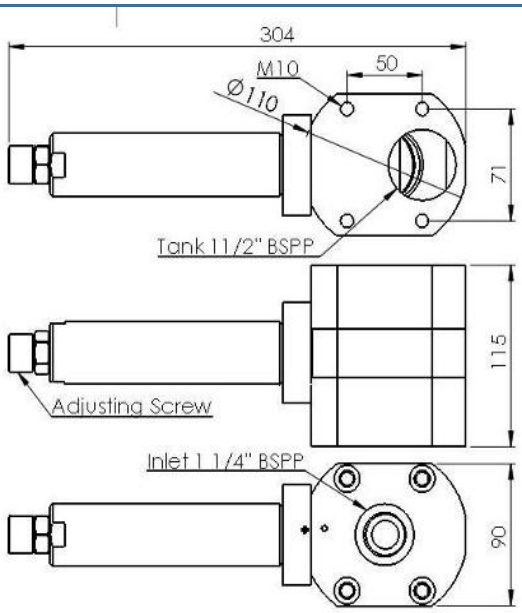




Maximum flow on the DN12 relief valve 110 l/min

207EXSW





Maximum flow on the DN16 relief valve 450 l/min

The pressure control valves are not "Safety Accessories" as defined in the Pressure Equipment Directive 97/23/EC.

Set range and relieving capacity must suit the protected circuit. DN12 capacity: 0-120 L/min. Chart flow ticks have unequal numeric steps at equal spacing; do not interpolate by horizontal distance. It is not a zero-leakage isolation valve; verify return pressure and allowable override. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Relief Valves V.01.25.pdf, p. 3. Manufacturer labels and dimensions retained.

Valve Ordering Code

Selection data & performance | Manufacturer technical reference.



PRESSURE RATING	
*	(Bar)
1	<30
2	Up to 160
3	>60* (*Relief Valve Only)
4	25-100* (*Relief Valve Only)

MOUNTING	
*	
.....	Stand Alone
M	Janus Manifold
C	CETOP-3

VALVE TYPE	
*	
01	4/2
02	4/3 Closed Centre
03	3/2 Spring to P Closed
04	Priority Flow Control
05	Relief (Inline)
06	4/3 Closed P
07	Pressure Control (Janus)
08	3/2 Spring to P Open
09	Pressure Reducing
10	Check
11	PO Check Single Port 3:1
12	4/3 Open Centre
13	PO Check Double Port 3:1
14	4/3 P to A B at centre
15	Flow Divider
16	Pressure Compensated Flow Control
17	Solenoid Operated Un-Loader
18	- - - -
19	PO Check Single Port 9:1
20	- - - -
21	2/2 Normally Open
22	PO Check Double Port 9:1
23	Electronic Flow Control
24	- - - -
25	Electronic Pressure Control
26	2/2 Normally Closed
XX	Assembly

PORT ORIFICE	
*	
A	3
B	6
C	9
D	12
E	16
F	25
G	38

ACTUATION / RETURN	
*	
A	Solenoid 12VDC (No Card)
B	Solenoid 24VDC (Card Required)
C	Solenoid 48VDC (Card Required)
D	Detent
E	- - - -
F	Air Pilot
G	Manual
H	Flow Controlled
J	Water Pilot
K	- - - -
L	Oil Pilot
M	- - - -
N	24VDC Proportional
O	24VDC (No Card)
P	48VDC (No Card)
R	Stepper Motor Control
S	Spring* (*Return Only)
X	Pressure

FLUID	
*	
O	Other
S	Sea Water
T	Tech. Water
W	Tap Water

Example:

A Valve with a system operating pressure of 140 Bar, 4/3 Closed P, 6mm Ports, Manifold Mounted, 24V DC Solenoid actuated spring return for tap water.

(Note: Janus Control Valves stating (Card Req'd) require an actuation card as additional item. (Please refer to actuation card data sheet)

Ordering Code = 2 M 0 6 B B S W

Use the code guide with the selected valve specification. Confirm every code segment, fluid, seal, actuation and port configuration in the supplier quotation. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Valve Ordering Code V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.

Zone Valves

Selection data & performance | Manufacturer technical reference.

A patented range of manifold mounted zone valves manufactured in 316 stainless steel for use within the high pressure water mist industry. The zone valve can be supplied fitted into a one to five station manifold, with pilot and solenoid operation.
Various porting configurations are available; please contact TWHC with your specific requirements.



Solenoid

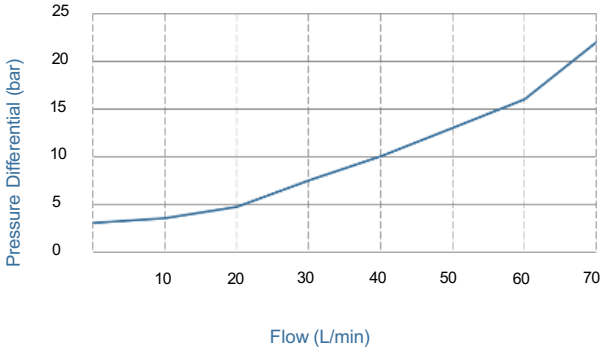
- Solenoid actuated 12 or 24 volts for independent zone operation.
- Coil power 32 watts.
- Maximum pressure 160 bar.
- 0-60 l/min per zone.
- Available in sizes DN3 , DN9 and DN25

Pilot

- Pilot actuated for independent zone operation.
- Pilot pressure 10 to 40 bar.
- Maximum pressure 160 bar.
- 0-60 l/min per zone.
- Available in sizes DN3 , DN9 and DN25
- ATEX approved



Fully opened pressure drop against flow rates



Specify open/closed duty, actuation supply, fail state, pressure drop and the fire-system control interface. Confirm approvals for the complete selected system. Water service removes hydraulic oil from the fluid circuit. Minimise unnecessary pressure loss; request configuration-specific approval evidence.

Zone Valves V.01.25.pdf, p. 1. Manufacturer labels and dimensions retained.



WaterQuest

SOLUTIONS

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